Self-purging bicycle tube valve

Abstract

A valve core for the inflation valve of a bicycle inner tube or other inflatable chamber holding liquid sealant has a core body (20), a poppet (40), a deflectable ring element such as a seat ring (30), and a return spring (50). In a closed state a sealing face (42) rests on a sealing edge (34), and a main passage (64) and a purge gap (60) are closed. Through a normal stroke the main passage opens while the purge gap remains closed. When the poppet moves beyond the normal stroke into over-travel, a purge shoulder (44) bears on the deflectable ring element and deflects it radially, opening the purge gap so that inflation gas flows along a separate purge flow path (62) across the surfaces on which sealant film accumulates. When the poppet returns, the deflectable ring element undergoes shape recovery, closes the purge gap and shears sealant film from the poppet.

Detailed Description

Definitions and conventions

Field

[0001] The present disclosure relates to inflation valves for inflatable chambers that hold liquid sealant, such as bicycle inner tubes and tubeless bicycle tyre set-ups, and to valve cores for such inflation valves. It relates in particular to a valve core having a deflectable ring element that is deflected radially when the poppet is displaced beyond its normal stroke, so that sealant film is removed from the surfaces of the valve core in the course of ordinary inflation, and to the seat ring, the methods of manufacture and assembly, the kit, the inflatable chamber, the wheel and the bicycle described herein.

Background

[0002] Liquid sealant carried in an inflatable chamber is drawn into the valve core of the inflation valve on each inflation event, by the pressure surge at the opening of the valve core and by capillary action along the poppet stem. Between inflation events the liquid sealant dries or cures on the surfaces of the valve core as sealant film. Sealant film on the sealing surfaces holds the poppet off its seat, producing leak-down of the inflatable chamber through the valve core, and sealant film that bridges the moving parts prevents the poppet from opening when the inflation device is applied, producing blockage. These are the two failure modes referred to in this specification as leak-down and blockage; both are defined in the test definitions below.

[0003] Existing responses to these failure modes act after the sealant film has formed or only delay its formation. A removable valve core can be unscrewed and cleared with a tool or replaced, which requires the user to act and, where a tool is inserted into the valve core, risks damage to it. A low-adhesion coating on the sealing surfaces slows the build-up of sealant film but does not remove a film once it has formed. An inflation valve of wider bore reduces the rate at which its passage is obstructed but does not clean its sealing surfaces. None of these removes sealant film from the sealing surfaces of the valve core as a consequence of the ordinary use of the inflation valve.

Summary

[0004] The valve core described herein removes sealant film from its own surfaces during ordinary inflation, without a tool and without any action by the user beyond inflating the inflatable chamber. The valve core comprises a core body, a deflectable ring element carried by the core body, and a poppet having a sealing face and a deflecting formation. In a closed state the sealing face rests on a sealing edge, which is provided by the deflectable ring element or by a fixed seat, and a purge gap associated with the deflectable ring element is closed. In a normal-open state the poppet is displaced from the closed state through a normal stroke, the sealing face is lifted from the sealing edge so that a main passage is open, and the purge gap remains closed. In a purge state the poppet is displaced beyond the normal stroke into over-travel, the deflecting formation bears on the deflectable ring element and deflects it radially, and the purge gap opens, so that a purge flow path distinct from the main passage carries inflation gas across the surfaces on which sealant film accumulates and into the inflatable chamber. When the poppet returns from over-travel the deflectable ring element undergoes shape

recovery, closing the purge gap, and in doing so moves across a surface of the poppet. The flow of inflation gas along the purge flow path is the air purge and the movement of the recovering deflectable ring element across the poppet is the snap-back wipe; the cleaning action of the valve core is the air purge, the snap-back wipe, or both.

[0005] Over-travel is produced by the pressure differential across the poppet exceeding an over-travel threshold that is higher than the crack pressure at which the normal stroke begins, so that the purge state is reached during inflation with an inflation device that raises the pressure differential above the over-travel threshold; in other embodiments over-travel is produced by mechanical depression of the poppet. In the illustrated embodiment the deflectable ring element is a seat ring that also provides the sealing edge, is formed from a superelastic alloy, and has a convoluted wall that converts the axial load applied by the deflecting formation into radial deflection. The seat ring, the deflecting formation, the threshold-setting arrangement, the purge gap and the purge flow path may each take the alternative forms described herein.

[0006] This specification also describes the seat ring as a component and as part of a kit, a method of manufacturing the seat ring, a method of assembling the valve core, a method of inflating an inflatable chamber, an inflation valve comprising the valve core, and an inflatable chamber, a wheel and a bicycle incorporating the inflation valve.

Brief description of the drawings

[0007] FIG. 1 is an axial section through the inflation valve of the illustrated embodiment, on a plane containing the poppet axis, in the closed state.

[0008] FIG. 2 is the section of FIG. 1 in the normal-open state.

[0009] FIG. 3 is the section of FIG. 1 in the purge state, showing the seat ring in radial deflection with its at-rest outline in dashed line, and showing the main passage and the purge flow path with arrows.

[0010] FIG. 4 is the section of FIG. 1 during shape recovery of the seat ring after removal of the inflation device, showing the snap-back wipe.

[0011] FIG. 5 shows the components of the inflation valve separated along the poppet axis in assembly order.

[0012] FIG. 6A is an enlarged detail of the region of the seat ring, the internal shoulder and the purge shoulder in the closed state, with the seat ring dimensions called out; FIG. 6B is the same detail in the purge state, with the purge gap opening called out.

[0013] FIG. 7 is a diagram of the states of the valve core and of the conditions governing the transitions between them.

[0014] FIGS. 8A to 8E illustrate manufacture and assembly: FIG. 8A the laser cutting of the seat ring from tube stock on a mandrel; FIG. 8B the shape-setting of the seat ring in a fixture; FIG. 8C the pressing of the seat ring into the core body against the internal shoulder; FIG. 8D retention of the seat ring by a crimped lip; and FIG. 8E retention of the seat ring by a retaining sleeve.

[0015] FIG. 9 is a section corresponding to FIG. 1 of an embodiment in which the seat ring is formed from a resilient polymer.

[0016] FIGS. 10A and 10B are sections of an embodiment having a fixed seat and a separate wiper ring, in the closed state and in the purge state respectively.

[0017] FIGS. 11A to 11C show alternative deflecting formations: a tapered purge shoulder, discrete lugs and a separate collar respectively.

[0018] FIGS. 12A to 12C show alternative wall geometries of the deflectable ring element: a frustoconical wall, a slotted wall and a cellular wall respectively.

[0019] FIGS. 13A and 13B show alternative purge flow paths: through purge ports in the wall of the core body, and along flutes in the core body bore, respectively.

[0020] FIGS. 14A and 14B show alternative threshold-setting arrangements: an auxiliary spring engaged after a lost-motion clearance, and a detent, respectively.

[0021] FIGS. 15A and 15B show schematically the mechanism in a Schrader-pattern valve core and in a non-removable inflation valve respectively.

Definitions and conventions

[0022] The definitions and conventions in this section apply throughout the specification and the claims. Where a defined term is also described with reference to an illustrated embodiment, the illustrated embodiment is one example of the defined term and does not limit it. The reference numerals used in this specification are those of the drawings, and the same numeral denotes the same component in every figure in which it appears.

General construction

[0023] As used herein, "comprising" and "comprises" are open transitional terms. A valve core, a seat ring, or any other element or combination that "comprises" a set of recited components or steps may include additional, unrecited components or steps, and is not limited to those recited.

[0024] Language such as "in some embodiments", "in other embodiments", "optionally" and "may" identifies a feature as preferred rather than essential to every embodiment described herein. The absence of such language for a given feature does not, by itself, make that feature essential; essentiality is determined from the description as a whole and, for a claimed combination, from the claim.

[0025] As used herein, "a" and "an" mean "one or more" unless the context clearly requires otherwise. For example, "a convolution" of the convoluted wall encompasses an embodiment having a single convolution and an embodiment having two or more convolutions, and "a purge shoulder" encompasses a purge shoulder that contacts the deflectable ring element at a single location or at two or more locations around its circumference.

[0026] Unless expressly qualified as "directly", terms describing the relationship between components, including "carried by", "coupled to", "on", "against" and "bears on", encompass both a direct connection or contact and an indirect connection or contact through one or more intervening components. For example, a purge shoulder "on the poppet" encompasses a purge shoulder integral with the poppet and a purge shoulder formed on a separate part fixed to the poppet.

[0027] Unless stated otherwise, a numerical range disclosed herein is inclusive of its stated endpoints, and disclosure of a range constitutes disclosure of every value and every narrower sub-range falling within it. Where a range is given herein for a parameter for which a single value is reported in the working examples, the range is the range within which the mechanism described is expected to operate, and values within the range other than the reported value are not represented as having been tested.

[0028] As used herein, "about" or "approximately", applied to a stated value, accounts for ordinary variation arising from manufacturing tolerance and measurement, consistent with the level of precision of the surrounding disclosure; it does not itself define a specific numerical tolerance unless one is separately stated. For example, "approximately 0.03 mm" describes the interference fit of the working example without asserting a measured tolerance band, and "nominally 55.8 wt%" describes the nickel content of the working example without asserting a measured compositional tolerance.

[0029] Where this specification describes a sequence of states or events, for example the closed state, the normal stroke, over-travel and the purge state, and shape recovery, that description identifies the pressure differential at which each state or event occurs and does not require that the states or events be discrete, mutually exclusive, or performed only in the order presented, except where the description or the context makes a particular order necessary; for example, over-travel necessarily follows completion of the normal stroke as the pressure differential rises.

[0030] Certain terms used herein combine a generic structural term with a statement of function, for example "deflectable ring element", "threshold-setting arrangement" (of which the second spring stage of the return spring is the species described in detail), "deflecting formation" (of which the purge shoulder is the species described in

detail) and "biasing member" (of which the return spring is the species described in detail). Each such term refers to any structure capable of performing the recited function, consistent with the corresponding structures described in this specification and their equivalents; this specification describes at least one corresponding structure for each such term so used, and the term is not limited to the material, form or manner of retention of that structure.

Orientation and directions

[0031] As used herein, the "supply side" of the poppet means the side coupled to the inflation device, and the "chamber side" of the poppet means the side open to the inflatable chamber. These terms replace orientation-dependent words, because the orientation of the inflation valve changes as the wheel turns. "Axial" means along the axis of the poppet, referred to herein as the poppet axis, and "radial" means perpendicular to the poppet axis. "Outward" means away from the poppet axis and "inward" means towards it. Movement of the poppet from the closed state through the normal stroke and into over-travel is referred to as movement in the opening direction. The deflectable ring element is "at rest" when it is not deflected by the purge shoulder.

Pressure differential, crack pressure, over-travel threshold and poppet travel

[0032] As used herein, the "pressure differential" means the supply-side pressure less the chamber-side pressure across the poppet. The "crack pressure" means the pressure differential at which the poppet begins its normal stroke, overcoming the first spring stage of the return spring. The "over-travel threshold" means the pressure differential above which the poppet moves beyond the normal stroke into over-travel, overcoming the second spring stage of the return spring or other threshold-setting arrangement. The over-travel threshold is higher than the crack pressure. In the illustrated embodiment the over-travel threshold is selected so that it is exceeded during inflation of the inflatable chamber with a floor pump, as in the working example described below.

[0033] As used herein, the "normal stroke" means the displacement of the poppet from the closed state, beginning at the crack pressure, over which the main passage is open and the purge gap remains closed. In the illustrated embodiment the purge shoulder is spaced from the deflectable ring element throughout the normal stroke; in other embodiments described below the purge shoulder contacts the deflectable ring element towards the end of the normal stroke without deflecting it so as to open the purge gap. "Over-travel" means movement of the poppet beyond the normal stroke, during which the purge shoulder bears on the deflectable ring element and deflects it radially so as to open the purge gap. In the illustrated embodiment over-travel occurs when the pressure differential exceeds the over-travel threshold; in the embodiments so described below it is produced by mechanical depression of the poppet. The normal stroke, over-travel and the purge state are defined by the position of the poppet and by the resulting condition of the deflectable ring element and the purge gap, whatever produces the displacement of the poppet. The "over-travel distance" means the axial distance the poppet moves beyond the end of the normal stroke during over-travel.

Operating states

[0034] The valve core has a closed state, a normal-open state and a purge state. In the illustrated embodiment the state is determined by the pressure differential. In the "closed state", which prevails when the pressure differential is below the crack pressure or when the lock nut is engaged, the sealing face of the poppet rests on the sealing edge, which is provided by the seat ring or, in embodiments having a fixed seat, by the fixed seat; the main passage is closed and the purge gap is closed. In the "normal-open state", which prevails when the pressure differential is at or above the crack pressure and does not exceed the over-travel threshold, the poppet is displaced from the closed state within the normal stroke, the sealing face is lifted from the sealing edge so that the main passage is open, the deflectable ring element is not deflected so as to open the purge gap, the purge shoulder in the illustrated embodiment being spaced from it, and the purge gap remains closed. In the "purge state", which prevails when the poppet is displaced beyond the normal stroke into over-travel, in the illustrated embodiment because the pressure differential exceeds the over-travel threshold, the purge shoulder bears on the deflectable ring element, the deflectable ring element is in radial deflection, the purge gap is open, and the air purge occurs while the main passage remains open. When the poppet returns from over-travel, in the illustrated embodiment when the pressure differential falls below the over-travel threshold, the deflectable ring element undergoes "shape recovery", meaning its return towards its at-rest shape, the purge gap closes, and the valve core returns through the normal-open state to the closed state as the pressure differential falls below the crack pressure.

Deflectable ring element, seat ring, wiper ring, fixed seat and convoluted wall

[0035] As used herein, a "deflectable ring element" means any annular element, carried by the valve core, that the purge shoulder deflects radially during over-travel and that undergoes shape recovery towards its at-rest form when the purge shoulder withdraws. In the embodiments having a purge gap, which include the illustrated embodiment, the radial deflection opens a purge flow path; in the wipe-only embodiment described below, in which the periphery of the deflectable ring element is sealed, the radial deflection and the shape recovery that follows it produce the snap-back wipe without opening a purge flow path. A deflectable ring element is a seat ring, where the same element also provides the sealing edge, or, in embodiments having a separate fixed seat, a wiper ring. The deflectable ring element is described below with reference to the seat ring of the illustrated embodiment; the term is not limited to the material, the wall form or the manner of retention of that seat ring.

[0036] As used herein, a "seat ring" means a deflectable ring element that both provides a sealing edge on which the sealing face of the poppet rests in the closed state, and is deflected by the purge shoulder during over-travel to open the purge gap. In the illustrated embodiment the seat ring is formed from a superelastic alloy and has a convoluted wall; in other embodiments it is formed from a resilient polymer. The "sealing edge" means the edge of the seat ring, or of the fixed seat, on which the sealing face rests in the closed state.

[0037] As used herein, a "wiper ring" means a deflectable ring element, separate from a fixed seat, positioned on the chamber side of the fixed seat so that the purge shoulder deflects it outward during over-travel to open the purge gap, while the fixed seat provides the sealing edge. A "fixed seat" means a component of the valve core that does not deflect during over-travel and that provides the sealing edge on which the sealing face of the poppet rests, used together with a wiper ring.

[0038] As used herein, a "convoluted wall" means a wall of the seat ring formed with one or more shallow convolutions, which converts axial load applied to the seat ring into radial deflection of the seat ring. The number of convolutions means the count of convolutions in the convoluted wall; the illustrated embodiment has one convolution or two convolutions.

Purge gap, purge flow path, main passage and radial deflection

[0039] As used herein, the "purge gap" means a clearance, associated with the deflectable ring element, that is closed in the closed state and in the normal-open state and that is opened by radial deflection of the deflectable ring element in the purge state, through which the air purge flows. In the illustrated embodiment the purge gap is an annular radial clearance between the outer surface of the seat ring and the core body bore; in other embodiments the purge gap is bounded by another surface of the deflectable ring element, and the term is not limited to the location of the illustrated embodiment.

[0040] As used herein, the "purge flow path" means the path, distinct from the main passage, from the supply side through the purge gap to the inflatable chamber, along which the air purge travels. The "main passage" means the passage past the sealing face that carries inflation gas into the inflatable chamber in the normal-open state and in the purge state. The main passage and the purge gap are different flow paths: the main passage is opened by lifting of the sealing face from the sealing edge at the crack pressure, whereas the purge gap is opened only by radial deflection of the deflectable ring element in the purge state. The "purge gap opening" means the radial size of the purge gap when the deflectable ring element is fully deflected in the purge state.

[0041] As used herein, "radial deflection" means elastic deformation of the wall of the deflectable ring element, produced by the purge shoulder during over-travel, that opens the purge gap or, in the wipe-only embodiment, that is followed by the shape recovery which produces the snap-back wipe. Radial deflection is a deformation of the deflectable ring element while it remains retained in the valve core; it is not a bodily displacement of the element as a whole. The radial deflection may be outward, inward, or non-uniform around the circumference of the deflectable ring element. The sense of radial deflection that opens a given purge gap is the sense that moves the portion of the deflectable ring element bounding that purge gap away from the opposed surface against which it bounds the purge gap, as described in relation to the purge state.

Purge shoulder and deflecting formation

[0042] As used herein, a "purge shoulder" means the formation of the poppet that contacts the deflectable ring element only during over-travel and thereby deflects it radially to open the purge gap. In the illustrated embodiment the purge shoulder lies on the chamber side of the sealing face. In every embodiment the purge shoulder is positioned along the poppet so that, as the poppet moves in the opening direction, the purge shoulder moves towards the surface of the deflectable ring element on which it bears, and is spaced from that surface in the closed state by an axial clearance equal to the normal stroke. The purge shoulder is a deflecting formation, meaning a formation that moves with the poppet and that contacts the deflectable ring element only during over-travel; the term "purge shoulder" is defined by that function and is not limited to the annular form of the purge shoulder of the illustrated embodiment.

Return spring, spring stages and threshold-setting arrangement

[0043] As used herein, the "return spring" means the biasing member that biases the poppet towards the closed state. The "first spring stage" means the stage of the return spring having the lower rate, which resists the normal stroke and thereby sets the crack pressure. The "second spring stage" means the stage of the return spring having a higher rate than the first spring stage, which the pressure differential must overcome, once the normal stroke is complete, to produce over-travel, and which thereby sets the over-travel threshold. A "threshold-setting arrangement" means any arrangement of the valve core that sets the over-travel threshold; the return spring having a first spring stage and a second spring stage is the threshold-setting arrangement of the illustrated embodiment.

Cleaning actions, sealant and inflation terms

[0044] As used herein, "cleaning action" means an action by which the valve core removes sealant film from its surfaces without a tool or user action. The cleaning actions described herein are the air purge, the snap-back wipe, or both. The "air purge" means the flow of inflation gas, at the pressure of the inflation device, along the purge flow path during the purge state, across the surfaces of the valve core on which sealant film accumulates and into the inflatable chamber. The inflation gas may be air supplied by a floor pump or another inflation device, or another gas supplied by a compressed-gas inflation device, such as a compressed-CO2 inflation device supplying carbon dioxide as the inflation gas. The "snap-back wipe" means the movement of the deflectable ring element across a surface of the poppet as the deflectable ring element undergoes shape recovery when the pressure differential falls below the over-travel threshold, by which movement partially cured sealant film is sheared from that surface.

[0045] As used herein, "sealant film" means liquid sealant that has dried or cured, in whole or in part, on a surface of the valve core. "Liquid sealant" means a fluid, carried in the inflatable chamber, that is drawn into the valve core on inflation and that is capable of drying or curing on the surfaces of the valve core; in the tested embodiment the liquid sealant is an emulsion of latex containing ammonia and suspended particles, and the term is not limited to that chemistry. "Blockage" means a failure mode of the valve core in which sealant film prevents the poppet from opening within a defined response time of actuation by the inflation device.

[0046] As used herein, an "inflatable chamber" means the volume, inflated through the inflation valve and holding liquid sealant, of a bicycle inner tube or of a tubeless bicycle tyre set-up. An "inflation device" means the source of inflation gas coupled to the supply side, for example a floor pump with a pump head, or a compressed-gas inflation device. An "inflation event" means the period from coupling of an inflation device to the supply side until the inflation device is removed.

Materials terms

[0047] As used herein, a "superelastic alloy" means an alloy that recovers large elastic strains on unloading, at the service temperature of the inflation valve, by a reversible phase transformation between martensite and austenite; in the illustrated embodiment the superelastic alloy is a nickel-titanium alloy (NiTi) of nominally 55.8 wt% nickel content. The "austenite finish temperature" means the temperature above which the superelastic alloy of the deflectable ring element is fully austenitic and exhibits superelastic behaviour. The "martensite start temperature" means the temperature, on cooling the superelastic alloy of the deflectable ring element, below which martensite begins to form and superelastic recovery may be impaired. The austenite finish temperature and the martensite start temperature are determined by differential scanning calorimetry, or by an equivalent

standard method for determining the transformation temperatures of a shape-memory alloy, and apply only to a deflectable ring element formed from a superelastic alloy.

[0048] As used herein, a "resilient metal" means a metal which, formed into the deflectable ring element, is deflected radially by the purge shoulder during over-travel and undergoes shape recovery when the pressure differential falls; a superelastic alloy is the resilient metal of the illustrated embodiment, and stainless steel is the resilient metal of the wiper ring described below. A "resilient polymer" means a polymer which, formed into the deflectable ring element, is deflected radially by the purge shoulder during over-travel and undergoes shape recovery when the pressure differential falls; polyether ether ketone (PEEK) is the resilient polymer described below.

Dimensional conventions

[0049] As used herein, the "seat ring inner diameter" means the inner diameter of the seat ring in its at-rest condition, measured at a portion of the wall free of convolutions, and the "seat ring outer diameter" means the outer diameter of the seat ring in its at-rest condition, measured at a portion of the wall free of convolutions. The "seat ring wall thickness" means the radial thickness of the material forming the wall of the seat ring, measured at the convoluted wall where the seat ring has a convoluted wall and otherwise at the portion of the wall that undergoes radial deflection. The "seat ring free length" means the axial length of the seat ring in its unloaded, at-rest condition. The "convolution amplitude" means the radial excursion of a convolution of the convoluted wall, measured from the mean radius of the wall to the peak or trough of the convolution. The "purge gap opening" and the "over-travel distance" are measured radially and axially respectively, as defined above, with the poppet at the end of over-travel.

Test definitions

[0050] As used herein, an "inflation cycle" means one repetition of an inflation event, a hold period at the resulting pressure of the inflatable chamber, and deflation of the inflatable chamber; in the working example an inflation cycle is inflation to 6 bar with a floor pump, a hold of 24 h at 22 °C, and full deflation through the inflation valve. "Leak-down" means the loss of pressure from the inflatable chamber, through the valve core, over a hold period. "Inflation time" means the elapsed time, using a given inflation device, from coupling to the supply side until the inflatable chamber reaches a stated pressure. A "conventional valve core" means a valve core lacking a deflectable ring element and a purge gap as defined herein, used as a comparator in the working example.

Structure of the valve core in the closed state

[0051] FIG. 1 is an axial section through the inflation valve of the illustrated embodiment on a plane containing the poppet axis, in the closed state. FIG. 5 shows the same components separated along the poppet axis in assembly order. FIG. 6A is an enlarged detail of the region of the seat ring (30), the internal shoulder (24) and the purge shoulder (44) in the closed state. The structure is described in this section in the closed state; the operation of the valve core through the normal-open state, the purge state and shape recovery is described in the following section.

Inflation valve and valve stem

[0052] Referring to FIG. 1, the inflation valve comprises a valve stem (10) having a stem bore (12), and a valve core received in the stem bore (12). The valve stem (10) is fixed to the inflatable chamber, which is a bicycle inner tube or a tubeless bicycle tyre set-up holding liquid sealant. In the illustrated embodiment the valve stem (10) is a conventional Presta-pattern valve stem of brass with an outer diameter of 6 mm, and the stem bore (12) is threaded. The core body (20) of the valve core is threaded into the stem bore (12), so that the valve core is retained in the valve stem (10) and can be removed from it as a unit. The Presta-pattern valve stem is described by way of example; the valve core and the inflation valve are not limited to that pattern.

The valve core

[0053] The valve core comprises a core body (20), a seat ring (30) carried by the core body (20), a poppet (40) guided by the core body (20), and a return spring (50) biasing the poppet (40) towards the closed state. The seat ring (30) is the deflectable ring element of the illustrated embodiment. The purge gap (60) is defined between the

seat ring (30) and the core body (20), as described below. A lock nut (70) is provided on the supply side. In FIG. 5 the components are shown in assembly order: the lock nut (70), the valve stem (10), the core body (20), the seat ring (30), the poppet (40) with its sealing face (42) and purge shoulder (44), and the return spring (50) with its first spring stage (52) and second spring stage (54). The valve core is fitted to and removed from the valve stem (10) as a whole, and may be supplied separately from the valve stem (10) as a replacement valve core for fitting to an existing valve stem.

Core body

[0054] The core body (20) is a turned component having an external thread engaged in the stem bore (12), a core body bore (22) that receives the seat ring (30), and an internal shoulder (24) against which the seat ring (30) is seated so that the seat ring (30) is located axially in the core body (20). The core body (20) also guides the poppet (40) in its axial movement. In the illustrated embodiment the core body (20) is turned from free-machining brass CW614N. In other embodiments the core body (20) is turned from stainless steel, for example grade 303, for corrosion resistance in the presence of ammonia-bearing liquid sealant; the core body (20) is turned by ordinary machining in either material and no other special process is required. The core body (20) is not limited to those materials.

Seat ring

[0055] The seat ring (30) is an annular element received in the core body bore (22) and seated against the internal shoulder (24). It has a first axial end and a second axial end. It has a sealing edge (34) at the first axial end, on which the sealing face (42) of the poppet (40) is seatable and on which the sealing face (42) rests in the closed state, and a convoluted wall (32) between the first axial end and the second axial end. In the illustrated embodiment the convoluted wall (32) has one convolution (32a) or two convolutions (32a, 32b), each convolution being a shallow radial excursion of the wall; the convoluted wall (32) converts axial load applied to the seat ring (30) into radial deflection of the seat ring (30). The axial load is applied by the purge shoulder (44) at a load-receiving surface of the seat ring (30) and is reacted by the internal shoulder (24), the convoluted wall (32) lying between the load-receiving surface and the end of the seat ring (30) that is seated against the internal shoulder (24), as described in relation to the purge state. The seat ring (30) is thin-walled relative to the core body (20), and its dimensions and their ranges are given below with reference to FIG. 6.

[0056] In the illustrated embodiment the seat ring (30) is formed from a superelastic alloy, namely a nickel-titanium alloy of nominally 55.8 wt% nickel content, having an austenite finish temperature specified below 5 °C so that the seat ring (30) is superelastic at riding temperatures. In other embodiments the seat ring (30) is formed from a resilient polymer, such as PEEK, with the same convoluted wall (32). The materials of the deflectable ring element, and the effect of the material on shape recovery and on the snap-back wipe, are described below.

[0057] The seat ring (30) is retained in the core body (20) by an interference fit, the interference being approximately 0.03 mm on diameter, and is pressed into the core body bore (22) until it is seated against the internal shoulder (24). In some embodiments the interference fit is provided over an axial band of the seat ring (30) adjacent the internal shoulder (24), and the remainder of the outer surface of the seat ring (30), including the portion of the convoluted wall (32) that undergoes radial deflection, is not in interference with the core body bore (22), so that this portion of the wall is free to move relative to the core body bore (22) during radial deflection. Over that remainder the outer surface of the seat ring (30) either lies in contact with the core body bore (22) without interference or stands at a clearance from it. In either case no flow passes along the outer surface of the seat ring (30) in the closed state or in the normal-open state, because the purge flow path (62) is closed at the purge gap (60), by the contact between the seat ring (30) and the core body bore (22) over the portion of the outer surface that bounds the purge gap (60) or by the ring retention at the axial band of the interference fit, and is opened only by the radial deflection of the seat ring (30) in the purge state. The pressure-retaining seal between the supply side and the chamber side is in every case formed at the sealing edge (34) by the sealing face (42). FIG. 6A shows the interference fit region between the seat ring (30) and the internal shoulder (24) and, distinct from it, the region in which the purge gap (60) is located. Other ways of retaining the seat ring (30) in the core body (20) are described below.

Poppet

[0058] The poppet (40) has a poppet stem (41), which is guided by the core body (20) and along which liquid sealant is drawn by capillary action during inflation, a sealing face (42), and a purge shoulder (44). In the illustrated embodiment the poppet (40) is of brass and the sealing face (42) is conical. In the closed state the sealing face (42) rests on the sealing edge (34) of the seat ring (30), and the seal between the supply side and the chamber side is formed at the sealing edge (34) by the sealing face (42); the closed-state seal is formed between the sealing face (42) and the sealing edge (34) without any elastomeric element in the illustrated embodiment. Other forms and materials of the sealing face (42) are described below.

[0059] The purge shoulder (44) is a formation of the poppet (40) axially spaced from the sealing face (42); in the illustrated embodiment it lies on the chamber side of the sealing face (42) and is an annular shoulder integral with the poppet (40). The purge shoulder (44) is positioned along the poppet (40) so that, as the poppet (40) moves in the opening direction, the purge shoulder (44) moves towards the load-receiving surface of the seat ring (30) on which it bears in over-travel, the load-receiving surface facing the purge shoulder (44) across an axial clearance. In the closed state the purge shoulder (44) is spaced axially from the seat ring (30) (FIG. 1, FIG. 6A) by an axial clearance equal to the normal stroke, 0.6 mm in the prototype, so that the purge shoulder (44) remains spaced from the seat ring (30) throughout the normal stroke and contacts the seat ring (30) only during over-travel. The purge shoulder (44) therefore takes no part in sealing and no part in the normal stroke; its only function is to bear on the load-receiving surface of the seat ring (30) during over-travel and to deflect the seat ring (30) radially. The load-receiving surface is the end face of the seat ring (30) at the end remote from the sealing edge (34), a convolution of the convoluted wall (32), or, where the deflecting formation is the sealing face (42) or a continuation of it, the region of the sealing edge (34), as described below among the alternative deflecting formations. Other forms of deflecting formation are described below.

Return spring

[0060] The return spring (50) biases the poppet (40) towards the closed state and has a first spring stage (52) and a second spring stage (54). The first spring stage (52) has the lower rate; it resists the normal stroke and sets the crack pressure. The second spring stage (54) has a higher rate than the first spring stage (52); it comes into action when the normal stroke is complete, it must be overcome by the pressure differential to produce over-travel, and it thereby sets the over-travel threshold. In the illustrated embodiment the first spring stage (52) and the second spring stage (54) are stages of a single return spring (50) of stainless steel music wire. The mechanism does not depend on the material of the return spring (50). The return spring (50) with its first spring stage (52) and second spring stage (54) is the threshold-setting arrangement of the illustrated embodiment; other threshold-setting arrangements are described below.

Purge gap in the closed state

[0061] In the illustrated embodiment the purge gap (60) is an annular radial clearance between the outer surface of the seat ring (30) and the core body bore (22) (FIG. 1, FIG. 6A). In the closed state the purge gap (60) is closed, and it remains closed in the normal-open state; it is opened only by radial deflection of the seat ring (30) when the purge shoulder (44) bears on the seat ring (30) during over-travel. A purge gap (60) so located is opened by movement of the portion of the outer surface of the seat ring (30) that bounds it away from the core body bore (22), as described in relation to the purge state and to the alternative locations of the purge gap. In the closed state the purge flow path (62) is closed at the purge gap (60), by contact between the seat ring (30) and the core body bore (22) over the portion of the outer surface that bounds the purge gap (60) or by the ring retention, and the pressure-retaining seal between the supply side and the chamber side is formed at the sealing edge (34) by the sealing face (42). The purge gap (60) is a flow path distinct from the main passage (64): the main passage (64) is the passage past the sealing face (42), which opens at the crack pressure, whereas the purge gap (60) is closed at the crack pressure and throughout the normal stroke and opens only in the purge state. The purge gap (60) and the purge flow path (62) through it are described in the purge state, and their location in other embodiments, below.

Lock nut

[0062] The lock nut (70) is a conventional Presta-pattern lock nut. When engaged by hand it holds the poppet (40) in the closed state irrespective of the pressure differential, for storage and transport. While the lock nut (70) is engaged the sealing face (42) rests on the sealing edge (34), the purge shoulder (44) is spaced from the seat ring

(30), and the seat ring (30) is at rest and undeflected; the lock nut (70) is therefore the preferred means of holding the inflation valve closed during storage and transport, rather than any accessory that bears on the poppet (40) in the opening direction. The lock nut (70) is a conventional component and takes no part in the cleaning actions.

Operation: the states of the valve core

[0063] The valve core operates in a closed state, a normal-open state and a purge state, and passes through a recovery from the purge state back to the closed state. The state that the valve core occupies is determined by the pressure differential across the poppet (40), that is by the pressure on the supply side less the pressure on the chamber side, acting against the return spring (50). FIG. 7 shows the sequence of states as a diagram, with the condition that governs each transition marked on the corresponding arrow. Control of the valve core is entirely mechanical; no electrical or electronic control is present.

[0064] In the closed state (FIG. 1) the pressure differential is below the crack pressure, the sealing face (42) of the poppet (40) rests on the sealing edge (34) of the seat ring (30), the main passage (64) is closed and the purge gap (60) is closed. In the normal-open state (FIG. 2) the pressure differential is at or above the crack pressure and below the over-travel threshold; the poppet (40) is displaced from the closed state by the normal stroke, the main passage (64) is open, the purge shoulder (44) is spaced from the seat ring (30), and the purge gap (60) remains closed. In the purge state (FIG. 3) the pressure differential exceeds the over-travel threshold, the over-travel threshold being higher than the crack pressure; the poppet (40) is in over-travel beyond the normal stroke, the purge shoulder (44) bears on the seat ring (30) and deflects it radially, and the purge gap (60), which is a flow path separate from the main passage (64), is open. The purge gap (60) is therefore closed in the closed state and in the normal-open state, and is open only in the purge state.

[0065] Recovery follows the purge state when the pressure differential falls below the over-travel threshold, as occurs when the inflation device is removed from the supply side. The return spring (50) returns the poppet (40) towards the closed state, the seat ring (30) undergoes shape recovery, the snap-back wipe occurs and the purge gap (60) closes (FIG. 4). The lock nut (70) may be tightened by hand to hold the poppet (40) in the closed state irrespective of the pressure differential, for storage and transport; FIG. 7 shows this as the locked condition. The lock nut (70) holds the poppet (40) in the closed state and not in over-travel, so that the seat ring (30) is not held in radial deflection while the inflation valve is stored.

[0066] The main passage (64) and the purge gap (60) are distinct flow paths. The main passage (64) is the passage past the sealing face (42) that carries inflation gas into the inflatable chamber whenever the sealing face (42) is lifted from the sealing edge (34), that is in the normal-open state and in the purge state. The purge gap (60) is a passage bounded at least in part by a surface of the seat ring (30); it is closed while the seat ring (30) is in its at-rest form and is opened only by the radial deflection of the seat ring (30). Inflation gas passing through the purge gap (60) follows the purge flow path (62), which is described below.

The crack pressure and the normal stroke

[0067] When an inflation device is coupled to the supply side and the pressure differential reaches the crack pressure, the force on the poppet (40) overcomes the first spring stage (52) of the return spring (50) and the poppet (40) moves along the poppet axis towards the chamber side, lifting the sealing face (42) from the sealing edge (34). The crack pressure is set by the first spring stage (52), which is the lower-rate stage of the return spring (50) and which alone resists the poppet (40) during the normal stroke. The normal stroke is the displacement of the poppet (40) from the closed state up to the position at which the purge shoulder (44) would first contact the seat ring (30). In the prototype the crack pressure is 1.0 bar and the normal stroke is 0.6 mm; the range of the crack pressure is given below together with the other operating parameters.

[0068] Throughout the normal stroke the purge shoulder (44) of the illustrated embodiment remains axially spaced from the seat ring (30) by an axial clearance, shown in FIG. 2 as the residual clearance between the purge shoulder (44) and the seat ring (30) at the end of the normal stroke. The seat ring (30) is not loaded by the poppet (40) during the normal stroke, retains its at-rest form, and the purge gap (60) remains closed. In embodiments in which the threshold-setting arrangement is the stiffness of the deflectable ring element, described below, the purge shoulder (44) contacts the seat ring (30) at the end of the normal stroke and loads it elastically without opening the purge gap (60), and the valve core remains in the normal-open state until the

pressure differential exceeds the over-travel threshold. Inflation gas flows through the main passage (64) only. Until the over-travel threshold is reached, therefore, the valve core admits inflation gas and seals in the same manner as a conventional valve core, the seat ring (30) serving only to seat the sealing face (42).

The over-travel threshold and over-travel

[0069] When the pressure differential exceeds the over-travel threshold, the poppet (40) moves beyond the end of the normal stroke into over-travel. The over-travel threshold is set by the second spring stage (54) of the return spring (50), which has a higher rate than the first spring stage (52) and which the poppet (40) engages at the end of the normal stroke. In the prototype the over-travel threshold is 3.5 bar and the over-travel distance is 0.5 mm. The normal stroke and the over-travel are two successive ranges of movement of the one poppet (40) along the same axis, distinguished by the pressure differential at which each occurs and by whether the purge shoulder (44) bears on the seat ring (30); no second closure member and no second seating surface is involved, and the seat ring (30) remains the valve seat member throughout.

[0070] The relationship between the pressure differential and the movement of the poppet (40) is as follows, and applies to every threshold-setting arrangement described herein. The force urging the poppet (40) in the opening direction is the pressure differential multiplied by the effective area of the poppet (40) on which the pressure differential acts, which in the closed state is the area bounded by the sealing edge (34). The crack pressure is the pressure differential at which that force equals the preload of the first spring stage (52), and is set by that preload. The over-travel threshold is the pressure differential at which that force equals the force of the return spring (50) at the end of the normal stroke together with the additional force needed to engage the second spring stage (54), or the other threshold-setting arrangement, and to begin deflecting the seat ring (30); it is set by the engagement point and rate of the second spring stage (54), or by the corresponding property of the other threshold-setting arrangement, together with the stiffness of the seat ring (30) where that contributes. Both are verified on the assembled valve core by measuring the pressure differential at which the poppet (40) lifts from the sealing edge (34) and the pressure differential at which the purge shoulder (44) contacts, or begins to deflect, the seat ring (30). The crack pressure and the over-travel threshold of the worked example are given below; the spring rates and preloads that produce them follow from these relationships and from the effective area of the poppet (40).

[0071] The over-travel distance is the axial distance that the poppet (40) moves beyond the end of the normal stroke. In some embodiments the over-travel distance is limited by an over-travel stop in the core body (20) that the poppet (40) meets at the end of the over-travel; in other embodiments it is limited by the second spring stage (54) reaching its solid length, or by the resistance of the seat ring (30) to further radial deflection.

[0072] The second spring stage (54) is one species of threshold-setting arrangement. In the illustrated embodiment the return spring (50) is a single compression spring having the first spring stage (52) and the second spring stage (54). Once the purge shoulder (44) bears on the seat ring (30), the resistance of the seat ring (30) to radial deflection adds to the force that the pressure differential must overcome, so that the return spring (50) and the seat ring (30) may each contribute to the over-travel threshold. Other threshold-setting arrangements, in which the step in resisting force at the end of the normal stroke is provided by a separate auxiliary spring engaged after a lost-motion clearance, by a detent, or by the seat ring (30) alone with a single-rate return spring (50), are described below with reference to FIG. 14.

[0073] The over-travel threshold is higher than the crack pressure, so that there is a normal-open state in which the valve core admits inflation gas without entering the purge state, and the seat ring (30) is deflected only when the inflation device raises the pressure differential above the over-travel threshold. In the prototype the crack pressure is 1.0 bar and the over-travel threshold is 3.5 bar. In the test described in Example 1, a floor pump inflating an inner tube from fully deflated to 6 bar produced over-travel on every inflation event. In some embodiments the over-travel threshold is set close to the crack pressure, so that the purge state is reached on substantially every opening of the valve core and the normal stroke is short; in such embodiments the second spring stage (54) may be omitted and the return spring (50) has a single rate. An embodiment in which the purge state is reached on every opening has not been built or tested.

The purge state

[0074] When the poppet (40) enters over-travel, the purge shoulder (44) comes into contact with the load-receiving surface of the seat ring (30) and bears on it with a force that rises with the pressure differential. The seat ring (30) is retained in the core body (20) against the internal shoulder (24) at one of its axial ends, and the load-receiving surface lies at the other axial end of the seat ring (30) or on the convoluted wall (32) between the ends, so that the axial load applied by the purge shoulder (44) is reacted by the internal shoulder (24) and compresses the convoluted wall (32) between the load-receiving surface and the seated end; the convoluted wall (32) converts that axial load into radial deflection of the wall. The axial order of the parts along the poppet axis is accordingly: the internal shoulder (24) and the seated end of the seat ring (30); the convoluted wall (32); the load-receiving surface; and, in the closed state, the purge shoulder (44), spaced from the load-receiving surface by the normal stroke and approaching it as the poppet (40) moves in the opening direction. The sealing edge (34) lies at whichever of the two axial ends of the seat ring (30) is the first axial end, and the sealing face (42) rests on it, in the closed state, from the side towards which the poppet (40) moves in the opening direction. In the illustrated embodiment the radial deflection is an elastic deformation of the wall of the seat ring (30) while the seat ring (30) remains retained by the core body (20) at the internal shoulder (24), rather than a bodily movement of the seat ring (30) relative to the core body (20).

[0075] The radial deflection opens the purge gap (60). The purge gap (60) is closed while the seat ring (30) is in its at-rest form, and opens when the radial deflection moves the portion of the seat ring (30) that bounds the purge gap (60) away from the opposed surface against which it bounds the purge gap (60). FIG. 3 shows the seat ring (30) in radial deflection, its at-rest outline repeated in dashed line, with the purge gap (60) open; FIG. 6B shows the same region enlarged, with the purge gap opening marked. In the embodiment of FIGS. 1 to 4 the purge gap (60) is located at the outer surface of the seat ring (30), between that surface and the core body bore (22), so that the purge gap (60) opens when the portion of the outer surface that bounds it moves away from the core body bore (22); that movement is an inward movement of the bounding portion, produced by the change in depth of a convolution of the convoluted wall (32) under the axial load, or a non-uniform deflection of the seat ring (30) around its circumference in which the bounding portion moves away from the core body bore (22) over part of the circumference, as described below in relation to the sense of the radial deflection and to the alternative locations of the purge gap. As described above, the interference fit between the seat ring (30) and the core body (20) may be confined to an axial band at the internal shoulder (24) so that the remainder of the outer surface of the seat ring (30) is free to move relative to the core body bore (22) (FIG. 6A). The purge gap opening is the radial size of the purge gap (60) when the seat ring (30) is fully deflected in the purge state; in the prototype the purge gap opening is 0.15 mm, a design value that was not measured directly during the bench test, and the sense of movement of the bounding portion of the seat ring (30) of the prototype was likewise not measured.

[0076] The sense of the radial deflection that opens the purge gap (60) depends on which surfaces bound the purge gap (60), and the rule is the same in every embodiment: the portion of the deflectable ring element that bounds the purge gap (60) moves away from the opposed surface. Where the purge gap (60) lies between the outer surface of the deflectable ring element and the core body bore (22) or the stem bore (12), the bounding portion moves inward, for example by flattening of an outwardly convex convolution or by deepening of an inwardly concave convolution under the axial load, or the deflection is non-uniform around the circumference so that the bounding portion moves away from the core body bore (22) over part of the circumference. Where the purge gap (60) lies between the inner surface of the deflectable ring element and the poppet (40), the bounding portion moves outward. Where the purge gap (60) is an opening through the wall of the deflectable ring element, the opening is widened or uncovered by the strain of the wall under the load of the purge shoulder (44), whether the wall moves outward, as where the fingers of a slotted wall (332) splay, or inward. An ovalising deflection, produced where the purge shoulder (44) contacts the deflectable ring element at discrete points around its circumference, moves the wall outward at the points of contact and inward between them, and so opens a purge gap (60) at the outer surface over the arcs between the points of contact and at the inner surface over the arcs at the points of contact. Every statement herein that the deflectable ring element is deflected radially outward is to be read with this rule: outward deflection opens a purge gap (60) at the inner surface or through the wall of the deflectable ring element, and a purge gap (60) at the outer surface is opened by inward or non-uniform movement of its bounding portion. The locations at which the purge gap (60) may be formed are described below with reference to FIG. 10 and FIG. 13.

[0077] In embodiments having a fixed seat (130) and a separate wiper ring (132) (FIG. 10), the sealing face (42) seats on the fixed seat (130) in the closed state, and in the purge state the purge shoulder (44) bears on and

deflects the wiper ring (132), which is the deflectable ring element of those embodiments; the purge gap (60) is then bounded by a surface of the wiper ring (132) and is opened by its radial deflection. Everything said in this section of the seat ring (30) as the element deflected by the purge shoulder (44) applies to the wiper ring (132) in those embodiments.

The purge flow path and the air purge

[0078] With the purge gap (60) open and inflation gas being supplied to the supply side, inflation gas flows from the supply side into the purge gap (60), along the purge flow path (62) past the seat ring (30) to the chamber side of the seat ring (30) and into the inflatable chamber, at the same time as inflation gas flows through the main passage (64). FIG. 3 shows both flows with arrows. In the embodiment of FIGS. 1 to 4 the purge flow path (62) enters the purge gap (60) at the end of the purge gap (60) that is open to the supply side, runs along the outer surface of the seat ring (30) over the portion of the convoluted wall (32) that undergoes radial deflection, and leaves the purge gap (60) at its other end by a route past the ring retention and the internal shoulder (24). That route is provided in the manner described below with reference to FIG. 13, by one or more purge ports (262) through the wall of the core body (20) leading to the clearance between the core body (20) and the stem bore (12), or by one or more flutes (362) in the core body bore (22) that communicate with the chamber side past the axial band of the ring retention; FIG. 3 shows the route schematically. The purge flow path (62) crosses the surfaces of the valve core on which sealant film accumulates, in the illustrated embodiment the outer surface of the seat ring (30) and the adjoining region of the core body bore (22), and the inflation gas carries sealant film from those surfaces into the inflatable chamber. This flow is the air purge.

[0079] The air purge occurs at the pressure of the inflation device, in the direction of inflation, so that sealant film removed by the air purge is carried away from the valve core into the inflatable chamber and not towards the inflation device. The air purge continues for as long as the pressure differential remains above the over-travel threshold and inflation gas is supplied, and ends when the pressure differential falls and the purge gap (60) closes.

[0080] The inflation gas of the air purge is any gas supplied by the inflation device: air from a floor pump, a hand pump or a compressor, or carbon dioxide from a compressed-CO_2 inflation device. The air purge occurs with any inflation device that raises the pressure differential across the poppet (40) above the over-travel threshold. The over-travel threshold is accordingly chosen to lie above the crack pressure and within the pressure differential that the intended inflation device produces. In the test described in Example 1 the inflation device was a floor pump.

[0081] In the illustrated embodiment the purge flow path (62) runs alongside the main passage (64), from the supply side through the purge gap (60) to the chamber side of the seat ring (30), the route past the ring retention and the internal shoulder (24) being as stated above. The purge ports (262) and the flutes (362) by which that route may be provided, and the other forms of the purge flow path (62), are described below with reference to FIG. 13.

Removal of the inflation device: shape recovery and the snap-back wipe

[0082] When the inflation device is removed from the supply side, the pressure differential falls. The return spring (50) returns the poppet (40) towards the closed state, the load applied by the purge shoulder (44) to the seat ring (30) is removed, and the seat ring (30) undergoes shape recovery to its at-rest form, closing the purge gap (60). As it recovers, the seat ring (30) moves across the sealing face (42) of the poppet (40), or across an adjacent surface of the poppet (40), and shears from that surface liquid sealant that has partially cured on it. This movement of the recovering seat ring (30) across a surface of the poppet (40) is the snap-back wipe. FIG. 4 shows the seat ring (30) during shape recovery, with the direction of its movement across the sealing face (42) indicated by an arrow and the purge gap (60) closing. The sealing face (42) then re-seats on the sealing edge (34) and the valve core is again in the closed state.

[0083] The relative movement that produces the snap-back wipe is driven by the shape recovery of the seat ring (30), which is held in the core body (20), and not only by the return stroke of the poppet (40): the wall of the seat ring (30) returns radially across the surface of the poppet (40) as the poppet (40) passes it, so that the shape recovery of the seat ring (30) itself moves the seat ring (30) relative to the poppet (40). In the embodiment of

FIGS. 1 to 4 the element that performs the snap-back wipe is the seat ring (30) itself, whose sealing edge (34) seats the sealing face (42) in the closed state. The snap-back wipe therefore acts on the sealing face (42), which is the surface that seals, and no separate part that could itself accumulate sealant film is needed to perform it. The sealing face (42) and the sealing edge (34) are the surfaces found clean at the end of the test described in Example 1.

[0084] In the illustrated embodiment the surface of the poppet (40) that is wiped is the sealing face (42) or a surface of the poppet (40) adjacent to the sealing face (42), and the region of the seat ring (30) that moves across it includes the region of the sealing edge (34), as shown in FIG. 4.

[0085] The strength of the snap-back wipe depends on the speed and force of the shape recovery, which depend on the material of the seat ring (30). Where the seat ring (30) is formed from a superelastic alloy, the shape recovery is rapid and the recovery force is high, so that the recovering seat ring (30) shears partially cured sealant film from the surface of the poppet (40) rather than riding over it; in the prototype a snap is audible on removal of the inflation device. A superelastic alloy is preferred for the seat ring (30) for this reason. Where the seat ring (30) is formed from a resilient polymer such as PEEK, the shape recovery is slower and the recovery force lower, so that the snap-back wipe is weaker; the movement of the recovering polymer seat ring (30) across the surface of the poppet (40) is nonetheless a snap-back wipe as described herein, and the air purge acts independently of it. A soft sealing element that deforms under the sealing face (42) has, by contrast, the disadvantage that liquid sealant embeds itself in the deformed sealing surface as it cures instead of being sheared from the sealing face (42).

The two cleaning actions

[0086] The valve core thus cleans its sealing surfaces twice in each inflation event that reaches the purge state: once by the air purge while the purge gap (60) is open, and once by the snap-back wipe as the seat ring (30) recovers. Both cleaning actions are produced by the poppet (40) moving into and out of over-travel under the pressure differential applied by the inflation device; neither requires a tool, a separate actuator or any action by the user beyond inflating the inflatable chamber. The air purge acts on the surfaces along the purge flow path (62); the snap-back wipe acts on the surface of the poppet (40) that the recovering seat ring (30) crosses.

[0087] In the embodiment of FIGS. 1 to 4 the cleaning action comprises both the air purge and the snap-back wipe. The two actions are, however, independent of one another. The snap-back wipe occurs whenever a deflectable ring element held in the core body (20) is deflected by the purge shoulder (44) in over-travel and undergoes shape recovery when the pressure differential falls, whether or not inflation gas flows through the purge gap (60) while the deflectable ring element is deflected. The air purge occurs whenever the purge gap (60) is open while inflation gas is supplied, whether or not the recovering deflectable ring element wipes the poppet (40). Accordingly, in some embodiments the cleaning action is provided by the air purge and the snap-back wipe together; in some embodiments it is provided principally by the air purge, for example where the deflectable ring element is of a resilient polymer whose shape recovery produces only a weak snap-back wipe; and in some embodiments it is provided by the snap-back wipe alone, the deflectable ring element being sealed at its periphery so that no inflation gas flows past it while it is deflected, and the radial deflection in over-travel followed by the shape recovery on release producing the snap-back wipe without an air purge. In the wipe-only embodiment the deflectable ring element is still deflected radially by the purge shoulder (44) only in over-travel, is still held in the core body (20) while it deflects, and still moves across the sealing face (42) or an adjacent surface of the poppet (40) as it recovers; only the purge flow path is absent. The test described in Example 1 shows the combined effect of the two cleaning actions in the prototype; it does not apportion that effect between them. Neither an embodiment in which the cleaning action is the snap-back wipe alone nor an embodiment in which it is the air purge alone has been built or tested.

Over-travel by other means and purge during deflation

[0088] In the embodiments described above, over-travel is produced by the pressure differential exceeding the over-travel threshold. In some embodiments over-travel is produced, in addition or instead, by mechanical depression of the poppet (40): a depressor in the head of the inflation device, or in an adaptor fitted between the inflation device and the supply side, pushes the poppet (40) beyond the normal stroke into over-travel when the inflation device is coupled, whatever the pressure differential. In such embodiments the purge state lasts for as

long as the inflation device is coupled, and the second spring stage (54) may be omitted; the normal stroke then ends at the position at which the purge shoulder (44) begins to deflect the seat ring (30) so as to open the purge gap (60), and the purge state is the state in which the poppet (40) is displaced beyond that position and the purge gap (60) is open, however the displacement is produced. In some embodiments the poppet (40) is pressed into over-travel by hand, with the lock nut (70) released. None of these mechanically actuated embodiments has been built or tested.

[0089] The purge state may also be reached during deflation of the inflatable chamber. When the poppet (40) is pressed into over-travel while the inflatable chamber is pressurised and no inflation device is supplying gas, the purge gap (60) opens and gas from the inflatable chamber flows through the purge gap (60) in the direction from the chamber side to the supply side, and may carry sealant film out of the valve core towards the supply side. The purge flow path (62) may therefore carry flow in either direction. In some embodiments the purge state is reached during deflation as well as during inflation; in other embodiments it is reached during deflation only, each inflation event using the normal stroke alone. Purge during deflation has not been tested.

Inflation events and inflation cycles

[0090] An inflation event is the period from coupling of an inflation device to the supply side until its removal. Within one inflation event that reaches the purge state, the valve core passes from the closed state through the normal-open state to the purge state as the pressure differential rises past the crack pressure and then past the over-travel threshold, remains in the purge state while the pressure differential exceeds the over-travel threshold, and returns through recovery to the closed state when the inflation device is removed. The two cleaning actions therefore occur once per inflation event: the air purge during the purge state and the snap-back wipe on recovery. In the test described in Example 1, in which each inflation event inflated an inner tube from fully deflated to 6 bar with a floor pump, the purge state was reached in every inflation event, as indicated by the snap audible on removal of the inflation device.

[0091] An inflation event in which the pressure differential does not exceed the over-travel threshold, for example where the inflation device produces a lower pressure differential, is completed in the normal-open state: the valve core admits inflation gas through the main passage (64) and closes on removal of the inflation device without entering the purge state and without the seat ring (30) being deflected. Whether a given inflation event reaches the purge state depends on the pressure differential that the inflation device produces across the poppet (40) during that event.

[0092] An inflation cycle is one repetition of an inflation event, a hold period at the resulting chamber pressure and deflation of the inflatable chamber. Over repeated inflation cycles, liquid sealant drawn into the valve core during one inflation event, and dried or partially cured during the following hold period, is acted on by the air purge and the snap-back wipe of the next inflation event that reaches the purge state. The behaviour of the prototype and of a conventional valve core over forty such inflation cycles is reported in Example 1 and Comparative Example 1.

Materials of the deflectable ring element

[0093] The deflectable ring element is defined above by what it does: it is carried by the valve core, it is deflected radially by the purge shoulder (44) during over-travel so as to open a purge flow path, and it undergoes shape recovery towards its at-rest form when the pressure differential falls. Any material that provides a recoverable radial deflection of the size needed to open the purge gap (60), and that returns the ring element to its at-rest form when the load of the purge shoulder (44) is removed, may be used. The material chosen determines the force and speed of the shape recovery, and therefore the strength of the snap-back wipe, and it determines the wall geometry needed to keep the strain in the wall within what the material recovers. The materials are described here for the seat ring (30) of the illustrated embodiment; they apply equally to the wiper ring (132) of the fixed seat and wiper ring embodiment described below, subject to the exceptions stated.

[0094] In the preferred embodiments the seat ring (30) is formed from a superelastic alloy. The seat ring (30) of the worked example is formed from a nickel-titanium alloy (NiTi) having a nickel content of nominally 55.8 wt%, from drawn tube stock. The alloy is specified to have an austenite finish temperature below 5 °C, so that the seat ring (30) is fully austenitic, and behaves superelastically, at the temperatures at which the inflation valve is used. A superelastic alloy recovers large strains on unloading by a reversible transformation between martensite and

austenite, so that a thin wall of the alloy can be deflected radially by an amount of the order of the purge gap opening and return to its at-rest form within the closing movement of the poppet (40), with a recovery that is fast and carries force. That recovery gives the seat ring (30) its strongest snap-back wipe: the recovering seat ring (30) shears partially cured sealant film from the surface of the poppet (40) it crosses rather than riding over it. A sealing element soft enough to deform under the sealing face (42), such as an elastomeric one, by contrast lets liquid sealant embed itself in the sealing surface, where neither the air purge nor a wipe reaches it; the superelastic alloy provides radial compliance without a soft sealing surface.

[0095] The austenite finish temperature of the worked example is an upper limit chosen so that the seat ring (30) is superelastic across riding temperatures. Below the martensite start temperature of the alloy, martensite forms on cooling and shape recovery may be impaired until the alloy is warmed again above its austenite finish temperature. The margin between the martensite start temperature and the lowest temperature the seat ring (30) reaches in service, including transient chilling by the expansion of gas from a compressed-gas inflation device, is therefore a design consideration. In some embodiments the alloy is selected, by its nickel content or by the shape-setting treatment described below, to have a lower austenite finish temperature and hence a wider margin. No value below 5 °C is specified herein, and the behaviour of the seat ring (30) when chilled by a compressed-gas inflation device was not tested. Where a superelastic alloy is used, the austenite finish temperature is a property of that alloy species; it has no counterpart for a resilient polymer or for a resilient metal that does not transform, and every reference herein to the austenite finish temperature is to be read as conditioned on the superelastic alloy species.

[0096] In some embodiments the superelastic alloy is a binary NiTi alloy, as in the worked example. In other embodiments it is a ternary NiTi alloy: a NiTiCu alloy, which gives a narrower transformation hysteresis and more stable behaviour over repeated cycles, or a NiTiFe or NiTiCr alloy, which gives a lower austenite finish temperature. The transformation temperatures and the plateau stress of the alloy chosen interact with the over-travel threshold in embodiments in which the stiffness of the deflectable ring element contributes to the threshold-setting arrangement, as described below. Copper-based superelastic alloys, such as Cu-Al-Mn and Cu-Al-Ni alloys, may also be used, but copper alloys are attacked by ammonia, so these alloys are contemplated only for use with an ammonia-free liquid sealant.

[0097] In other embodiments the deflectable ring element is formed from a resilient metal that recovers its shape elastically without a phase transformation. Examples are stainless steel of spring temper, cold-worked austenitic stainless steel, precipitation-hardened stainless steel such as 17-7PH, and beta-titanium alloys. Stainless steel of spring temper is the disclosed material of the wiper ring (132) of the fixed seat and wiper ring embodiment. The recoverable strain of such a metal is much lower than that of a superelastic alloy, so a deflectable ring element of resilient metal is given a thinner wall, a smaller radial deflection, or a slotted or cellular wall as described below, so that the purge gap opening is reached within the elastic range of the material. A resilient metal has no austenite finish temperature, and its recovery does not depend on temperature in the manner of a superelastic alloy.

[0098] In further embodiments the deflectable ring element is formed from a resilient polymer. The disclosed polymer is polyether ether ketone (PEEK). A seat ring (30) of PEEK having the same convoluted wall (32) as the superelastic seat ring provides radial compliance under the purge shoulder (44) at lower cost, but its shape recovery is slower and the force of the recovery lower, so that its snap-back wipe is weaker than that of the superelastic seat ring. The purge gap (60) and the air purge provide their cleaning action independently of the wipe, so the polymer seat ring retains the air purge in full. This embodiment is illustrated in FIG. 9 and is intended for a lower-cost product line. Other resilient polymers that resist the liquid sealant, such as polyphenylene sulphide (PPS) and polyetherimide (PEI), may be used; creep of the polymer under a sustained interference fit, and its resistance to an ammonia-bearing liquid sealant, are checked for the polymer chosen. Electropolishing, and the austenite finish temperature, have no application to a polymer deflectable ring element.

[0099] Whatever the material, the snap-back wipe is the movement of the recovering deflectable ring element across a surface of the poppet (40) as the pressure differential falls; the term carries no speed or force qualifier. The superelastic alloy is preferred because it gives the strongest wipe. Across the material classes described above the wipe force falls from superelastic alloy to resilient metal to resilient polymer, and the wall geometry and the over-travel distance are adapted accordingly, while the air purge is available with every material.

Wall geometry converting axial load into radial deflection

[0100] The purge shoulder (44) moves axially with the poppet (40). To open the purge gap (60), the deflectable ring element converts the axial load applied by the purge shoulder (44) into a radial deflection of the surface that bounds the purge gap (60). This axial-to-radial conversion geometry is the second defining feature of the deflectable ring element after its material, and the load chain is the same in every embodiment: the purge shoulder (44) bears on the deflectable ring element only in over-travel; the geometry of the ring element turns that axial load into radial deflection; the radial deflection opens the purge gap (60), which is closed in the closed state and the normal-open state; and the shape recovery of the ring element closes the purge gap (60) again when the load is removed. The wall geometry is chosen together with the material so that the purge gap opening is reached at the over-travel distance within the strain the material recovers. The geometry is loaded mechanically by the purge shoulder (44), at a defined location and at service temperature; for the mechanically actuated embodiments described in this section, no heating or other external actuation is involved in reaching the purge state by this route.

[0101] In the illustrated embodiment the geometry is a convoluted wall (32). The wall of the seat ring (30) between its first axial end and its second axial end is formed with one or more shallow convolutions, each a radial excursion of the wall of a convolution amplitude measured from the mean radius of the wall. The seat ring (30) is supported against the internal shoulder (24) at one of its axial ends and is loaded by the purge shoulder (44) at the load-receiving surface spaced axially from that end, so that the convoluted wall (32) between them is compressed axially. Axial load applied by the purge shoulder (44) shortens the wall and changes the depth of the convolutions, moving the wall radially. The number of convolutions in the disclosed embodiments is one or two; FIG. 6A shows a first convolution (32a) and, optionally, a second convolution (32b). A wall with three or more convolutions may be used; each additional convolution adds axial compliance and reduces the radial deflection obtained per convolution for a given load, and the number is limited to that at which the wall still moves radially rather than merely shortening. The profile of a convolution is not limited to a rounded wave; it may be a saw-tooth, or a helical or circumferential wave. Optionally, the root of each convolution is provided with a fillet radius, which reduces the concentration of strain at the root under repeated radial deflection.

[0102] The convolution amplitude, together with the seat ring wall thickness and the seat ring free length, sets the relationship between the axial load applied by the purge shoulder (44), the resulting radial deflection, and the strain in the wall. A shallow convolution in a thin wall gives a radial deflection of the order of the purge gap opening at a strain the superelastic alloy recovers fully, and the convoluted wall (32) is the preferred geometry for that reason.

[0103] Other axial-to-radial conversion geometries are shown in FIG. 12. In FIG. 12A the wall is a frustoconical wall (232): a plain wall with a conical flare, in which axial load applied at the narrow end increases the flare and moves the free end radially outward, with no convolution to form. In FIG. 12B the wall is a slotted wall (332): a short tubular wall having a continuous band at the end carrying the sealing edge (34) and axial slots cut from the other end, which divide the remainder of the wall into fingers (332a, 332b) that splay radially under the load of the purge shoulder (44). The slotted wall (332) gives a large radial deflection at low force and low strain, which suits a resilient metal, and the continuous band keeps the sealing edge (34) intact. In FIG. 12C the wall is a cellular wall (432): a pattern of diamond or chevron cells cut through the wall between two continuous end bands, the struts of which rotate under axial compression so that the wall enlarges radially. The cellular wall (432) is formed by the same laser cutting as the convoluted wall (32), and the sealing edge (34) is carried on one of the continuous end bands. The coupling between axial shortening and radial enlargement of a cellular wall depends on the cell geometry chosen.

[0104] In a further embodiment, not separately illustrated, the deflectable ring element is a conical annular disc: a thin frustoconical annulus held at one rim. Axial load from the purge shoulder (44) flattens the cone and moves the free rim radially, outward where the inner rim is held and inward where the outer rim is held. The radial movement per unit of axial stroke is small at shallow cone angles, and this form is used with a separate fixed seat (130) for sealing, in the manner of the fixed seat and wiper ring embodiment described below.

[0105] In a further embodiment the wall of the deflectable ring element is plain, with no conversion geometry of its own, and the radial deflection is produced by a tapered purge shoulder (244), FIG. 11A: a frustoconical flank of the poppet (40) enters the deflectable ring element during over-travel and expands it by direct radial contact with

its inner surface, as a mandrel expands a ring. The load on the ring element is then radial rather than axial. The flank slides on the inner surface of the ring element, with more friction and wear than the axial loading of the convoluted wall (32), and wipes that inner surface on the return stroke.

[0106] The frustoconical wall (232), the slotted wall (332), the cellular wall (432), the conical annular disc and the plain wall driven by a tapered purge shoulder (244) are alternatives that perform the same function as the convoluted wall (32), namely converting the movement of the poppet (40) in over-travel into radial deflection of the deflectable ring element, and any of them may be used with any of the materials described above, subject to the strain the material recovers. Each of these geometries may be applied to a seat ring (30) that also seals or, in the fixed seat and wiper ring embodiment, to a wiper ring (132) that does not seal.

Continuous, split and segmented forms, and the purge opening

[0107] The seat ring (30) of the illustrated embodiment is a continuous ring, and its sealing edge (34) is continuous around its circumference, so that the sealing face (42) of the poppet (40) seals on it in the closed state. Where the deflectable ring element also provides the sealing edge (34), the band of the ring that carries the sealing edge (34) remains continuous in every form described herein; the slotted wall (332) and the cellular wall (432) place their slots and cells away from that band.

[0108] The purge gap (60) of the illustrated embodiment is an annular clearance bounded at least in part by a surface of the seat ring (30), closed in the closed state and the normal-open state and opened by the radial deflection, as described above. The purge opening through which the air purge passes need not be an annular clearance at a surface of the ring element. In the slotted wall (332) and the cellular wall (432) the slots or cells pass through the wall, and in some embodiments they form the purge opening: they are closed at rest by contact of the wall with the surface of the core body bore (22) that faces them and are uncovered when the radial deflection moves the wall away from that surface, so that inflation gas passes through the wall of the deflectable ring element rather than around it. In such embodiments the ring element may be fitted in the core body bore (22) with no clearance at its outer surface at rest. A deflectable ring element that does not itself seal may also be segmented: a ring divided into fingers whose gaps open under the purge shoulder (44), or a split wiper ring having a single axial split that opens circumferentially under a tapered purge shoulder (244), the split being the purge opening. A split ring is not used as a one-piece seat ring (30), because the split would interrupt the sealing edge (34).

[0109] The purge opening is therefore, in the various embodiments, an annular gap at a surface of the deflectable ring element, one or more slits or apertures through its wall, or one or more gaps between the fingers of a segmented ring element. In each case it is closed in the closed state and the normal-open state and opened by the radial deflection in the purge state, and the deflectable ring element is continuous, slotted, cellular, split or segmented accordingly. Slots, cells and splits that pass through the wall can retain liquid sealant, which is a consideration in choosing between the forms.

Dimensions and parameter ranges

[0110] The dimensions and pressures given below are those of the superelastic seat ring (30) of the worked example, together with the ranges within which the mechanism is expected to operate. The relationships between the parameters govern, and the numbers follow from them. First, the over-travel threshold is higher than the crack pressure, so that the valve core opens and seals as a conventional valve core does until the purge state is reached, and the over-travel threshold lies within the pressure differential that the inflation device applies, so that the purge state is reached during an inflation event. Second, the purge gap (60) is closed at rest and remains closed throughout the normal stroke, the purge shoulder (44) being spaced from the deflectable ring element by at least the normal stroke at rest. Third, the radial deflection produced at the over-travel distance is at least the purge gap opening, and the strain it imposes on the wall is within what the material of the deflectable ring element recovers, so that the purge gap (60) closes again on shape recovery. Fourth, the seat ring inner diameter admits the poppet stem (41) and presents the sealing edge (34) to the sealing face (42), and the seat ring outer diameter fits the core body bore (22) with the interference fit at the internal shoulder (24). These relationships are what adapt the seat ring (30) to its function in the valve core and distinguish it from a wave-section ring used as a static seal between two members: the sealing edge (34) is a continuous edge at an axial end on which the sealing face (42) of a moving poppet (40) is seatable metal to metal; the convoluted wall (32) is dimensioned so that an axial

load applied by the poppet (40) at the load-receiving surface and reacted at the internal shoulder (24) produces, at the over-travel distance, a radial deflection at least equal to the purge gap opening at a strain the material recovers; and the recovery is a shape recovery on removal of that mechanical load at service temperature, not a recovery obtained by heating. A person skilled in the art who changes one parameter adjusts the others to preserve these relationships.

[0111] The seat ring inner diameter is from 2.0 mm to 2.6 mm; in some embodiments from 2.1 mm to 2.5 mm; in some embodiments from 2.2 mm to 2.4 mm; in the worked example 2.3 mm. The seat ring wall thickness, being the radial thickness of the material of the convoluted wall (32), is from 0.15 mm to 0.30 mm; in some embodiments from 0.16 mm to 0.24 mm; in some embodiments from 0.18 mm to 0.22 mm; in the worked example 0.20 mm. The seat ring free length, being the axial length of the seat ring (30) unloaded, is from 2.5 mm to 4.0 mm; in some embodiments from 2.6 mm to 3.5 mm; in some embodiments from 2.8 mm to 3.2 mm; in the worked example 3.0 mm. The convolution amplitude is from 0.08 mm to 0.20 mm; in some embodiments from 0.08 mm to 0.16 mm; in some embodiments from 0.11 mm to 0.14 mm; in the worked example 0.12 mm. The seat ring (30) of the worked example is formed from tube stock of 2.7 mm outer diameter and 0.20 mm wall thickness, so that its seat ring outer diameter is nominally 2.7 mm. These dimensions are called out in FIG. 6A.

[0112] The purge gap opening, being the radial size of the purge gap (60) when the deflectable ring element is fully deflected in the purge state and called out in FIG. 6B, is from 0.10 mm to 0.25 mm; in some embodiments from 0.11 mm to 0.20 mm; in some embodiments from 0.13 mm to 0.17 mm; in the worked example 0.15 mm. The over-travel distance, being the axial distance the poppet (40) moves beyond the end of the normal stroke, is from 0.4 mm to 0.9 mm; in some embodiments from 0.4 mm to 0.65 mm; in some embodiments from 0.44 mm to 0.56 mm; in the worked example 0.5 mm, beyond a normal stroke of 0.6 mm. The crack pressure is from 0.8 bar to 1.5 bar; in some embodiments from 0.8 bar to 1.2 bar; in some embodiments from 0.91 bar to 1.09 bar; in the worked example 1.0 bar. The over-travel threshold is from 2.5 bar to 5.0 bar; in some embodiments from 2.75 bar to 4.25 bar; in some embodiments from 3.2 bar to 3.8 bar; in the worked example 3.5 bar. The over-travel threshold of the worked example was reached in every inflation of the bench test, in which a floor pump inflated the inflatable chamber to 6 bar; whether a given hand pump or compressed-gas inflation device reaches a given over-travel threshold depends on the pressure differential that device applies, and the over-travel threshold is chosen for the inflation devices with which the valve core is to be used.

[0113] Of these values, only those of the worked example were built and tested, and the over-travel distance and purge gap opening of the worked example are design values that were not measured directly during the bench test. The ranges are stated for the seat ring (30) and the valve core generally; the worked example lying within them is the superelastic seat ring (30) with a convoluted wall (32), and no seat ring (30) of another material or wall geometry, and no wiper ring (132), has been built to verify that the same values apply to it. Where the deflectable ring element is formed from a resilient polymer or a resilient metal, has another of the axial-to-radial conversion geometries described above, or is a wiper ring (132), a value within the stated ranges is selected so that the relationships stated above hold for the stiffness and recoverable strain of that material and geometry; where those relationships cannot be satisfied within a stated range, the relationships govern and the value is adapted accordingly. The dimensions are likewise scaled to the bore available in a larger valve core or in an inflation valve in which the deflectable ring element is carried directly by the valve stem (10).

One-piece seat ring, or fixed seat with a separate wiper ring

[0114] In the illustrated embodiment one part, the seat ring (30), performs both functions: its sealing edge (34) is the surface on which the sealing face (42) of the poppet (40) rests in the closed state, and its convoluted wall (32) is the deflectable ring element that the purge shoulder (44) deflects in over-travel. The seat ring (30) is held in the core body (20) and does not move bodily; the purge gap (60) opens by elastic deformation of its wall while the seat ring (30) remains seated against the internal shoulder (24). In this one-piece form the part that seals is also the part that wipes: the shape recovery of the seat ring (30) moves it across the sealing face (42) or an adjacent surface of the poppet (40), so that sealant film is removed from the sealing surfaces by the seat ring (30) itself, with no wiping part in addition to it. The one-piece form has the fewest parts and the shortest axial length, and the inspection of the worked example after forty inflation cycles found the sealing edge (34) and the sealing face (42) clean.

[0115] In an alternative embodiment, FIGS. 10A and 10B, sealing and deflection are performed by separate parts. A fixed seat (130), which does not deflect during over-travel, is carried by the core body (20) and provides the sealing edge (34) on which the sealing face (42) rests in the closed state; in the disclosed form the fixed seat (130) is of brass. A separate wiper ring (132), which is the deflectable ring element of this embodiment, is positioned on the chamber side of the fixed seat (130), so that the purge shoulder (44) reaches it only in over-travel. In over-travel the purge shoulder (44) deflects the wiper ring (132) radially, in the disclosed form outward, opening the purge gap (60), which is bounded by a surface of the wiper ring (132), and the air purge flows along the purge flow path (62). Where the wiper ring (132) is deflected outward, the purge gap (60) that the outward deflection opens lies at the inner surface of the wiper ring (132), between the wiper ring (132) and the poppet (40), or is an opening through the wall of the wiper ring (132) or the split of a split wiper ring; where the purge gap (60) lies between the outer surface of the wiper ring (132) and the core body bore (22), as shown in FIG. 10B, it is opened by inward or non-uniform movement of the bounding portion of the wiper ring (132), as described above for the seat ring (30). When the pressure differential falls the wiper ring (132) recovers its shape and the purge gap (60) closes. In the disclosed form the wiper ring (132) is a thin ring of stainless steel of spring temper. In other embodiments the wiper ring (132) is formed from any of the materials described above for the deflectable ring element, and it has a plain wall, a convoluted wall (32) or another of the axial-to-radial conversion geometries described above, or it is a split wiper ring as described above. Where the recovering wiper ring (132) moves across a surface of the poppet (40) it also provides a snap-back wipe; where it does not, this embodiment provides the air purge alone. The fixed seat and wiper ring embodiment has one more part and more axial length than the one-piece seat ring (30), and may be easier to manufacture; the form, thickness and retention of the wiper ring (132) follow from the material and geometry selected as described above.

[0116] Both forms are species of the same deflectable ring element: the seat ring (30) where the ring element also carries the sealing edge (34), and the wiper ring (132) where a fixed seat (130) carries the sealing edge (34). The description of materials, wall geometry, purge opening and dimensions above applies to both, and every reference herein to the deflectable ring element includes both unless the context excludes one. The parameter ranges given above are stated for the deflectable ring element generally and were verified only for the one-piece superelastic seat ring (30), as stated in relation to the dimensions and parameter ranges.

Separate part, integral flexure section, or carried by the valve stem

[0117] In the illustrated embodiment the seat ring (30) is a part separate from the core body (20), received in the core body bore (22) and retained against the internal shoulder (24). The retention of a separate deflectable ring element in the core body (20), by the interference fit of the worked example or by the other forms of ring retention, is described below with the alternatives for the other parts.

[0118] In other embodiments the deflectable ring element is not a separate part but an integral flexure section of the core body (20): a thin-walled section of the core body (20), having a convoluted wall (32), a slotted wall (332) or another of the axial-to-radial conversion geometries described above, formed in one piece with the core body (20) by turning, moulding or pressing. There is then no separate seat ring (30) and no interference fit. The core body (20) is in that case formed from a material having the recoverable strain the section requires, such as a resilient polymer or stainless steel of spring temper; the free-machining brass of the illustrated core body (20) does not recover the strains involved and is not used for an integral flexure section. The functional definition of the deflectable ring element covers a ring element separate from the core body (20) and one integral with it.

[0119] In further embodiments the deflectable ring element is carried not by a removable core body (20) but directly by the valve stem (10) of the inflation valve, as the seat ring (730) of FIG. 15B, in an inflation valve having no separate valve core; the valve formats in which the mechanism may be used are described below.

Shape recovery

[0120] Shape recovery is the return of the deflectable ring element to its at-rest form when the pressure differential falls and the purge shoulder (44) withdraws, closing the purge gap (60); the snap-back wipe is the movement of the recovering ring element across a surface of the poppet (40) during that return. The seat ring (30) thus recovers elastically, that is undergoes shape recovery to its at-rest form, on removal of the axial load applied by the purge shoulder (44). In the illustrated embodiment the shape recovery is the superelastic recovery of the NiTi seat ring (30): on unloading, the martensite formed under stress in the deflected wall reverts to austenite and

the wall returns to its shape-set form, quickly and with force. In the resilient polymer seat ring (30) of FIG. 9 the shape recovery is the elastic recovery of the polymer, slower and with less force. In a deflectable ring element of resilient metal, including the stainless steel wiper ring (132), the shape recovery is the elastic recovery of the metal.

[0121] In some embodiments the shape recovery is assisted by the poppet (40) itself: a second formation of the poppet (40), on the side of the deflectable ring element opposite the purge shoulder (44), meets the ring element on the return movement of the poppet (40) and drives it back towards its at-rest form. Closing of the purge gap (60) then does not depend only on the material's own recovery, which guards against residual set of the ring element after many cycles and against loss of superelastic recovery when the ring element is chilled below its martensite start temperature, at the cost of additional contact and friction on the return.

[0122] Where a superelastic deflectable ring element has been chilled below its martensite start temperature while deflected, for example by the expansion of gas from a compressed-gas inflation device, and has not fully recovered on unloading, it recovers its shape when it is warmed again above its austenite finish temperature. This recovery on warming is a property of the alloy and is distinct from the thermally actuated purge described below, in which the purge gap (60) is deliberately opened by warming. The seat ring (30) of the illustrated embodiment is specified with an austenite finish temperature below 5 °C so that it is austenitic, and recovers superelastically, at ordinary riding temperatures without any warming. Whether chilling by a compressed-gas inflation device takes the seat ring (30) below its martensite start temperature was not tested.

[0123] The worked example completed forty inflation cycles without failure of the seat ring (30); its fatigue life beyond forty over-travel cycles was not tested. Under repeated radial deflection the strain in a convoluted wall (32) is concentrated at the roots of the convolutions. The fillet radius at the convolution root described above, and the use of two convolutions (32a, 32b) rather than one so that the radial deflection is shared between them, are design measures that may be used to reduce the strain at any one location.

Alternative locations of the purge gap

[0124] The purge gap (60) is defined herein by its function: it is a clearance bounded at least in part by a surface of the deflectable ring element, which is closed in the closed state and in the normal-open state, which is opened by radial deflection of the deflectable ring element in the purge state, and through which the air purge flows. The purge gap (60) is distinct from the main passage (64): the main passage (64) is opened by lifting of the sealing face (42) from the sealing edge (34) during the normal stroke, whereas the purge gap (60) remains closed throughout the normal stroke and opens only when the deflectable ring element is deflected. In the embodiment of FIG. 1 to FIG. 4 the purge gap (60) is located between the outer surface of the seat ring (30) and the core body bore (22). The surfaces that bound the purge gap (60), and the sense of the radial deflection that opens it, may however be chosen in any of the ways described in this section, each of which is a species of the same functional definition.

[0125] In a first arrangement, which is that of the embodiment of FIGS. 1 to 4, the purge gap (60) is bounded by the outer surface of the seat ring (30) and the core body bore (22). Such a purge gap (60) opens where the bounding portion of the outer surface of the seat ring (30) moves away from the core body bore (22). That movement may be an inward movement of the bounding portion, produced for example by flattening of an outwardly convex convolution of the convoluted wall (32) under axial load from the purge shoulder (44), or by deepening of an inwardly concave convolution under the same load; or it may be the non-uniform deflection described below, in which some arcs of the seat ring (30) move outward while the intervening arcs move inward, so that the purge gap (60) opens over the inwardly moving arcs. Where the seat ring (30) is retained by an interference fit confined to an axial band at the internal shoulder (24), as described with reference to FIG. 6, the remainder of the outer surface is free of interference with the core body bore (22) and is available to bound the purge gap (60). Where the end of the purge gap (60) at which the purge flow path (62) leaves it lies at the ring retention and the internal shoulder (24), the purge flow path (62) passes them through purge ports (262) or flutes (362), as described below.

[0126] In a second arrangement the purge gap (60) is bounded by the outer surface of the seat ring (30) and the stem bore (12), the wall of the core body (20) being provided with one or more windows at the axial position of the seat ring (30) through which the outer surface of the seat ring (30) is exposed to the stem bore (12). This

arrangement requires the same movement of the outer surface as the first arrangement, and the windows in the core body (20) form part of the purge flow path (62).

[0127] In a third arrangement the purge gap (60) is bounded by the inner surface of the seat ring (30) and the poppet stem (41), at a portion of the inner surface spaced axially from the sealing edge (34). Such a purge gap (60) is opened by outward radial deflection of the seat ring (30), and the air purge passes along the poppet stem (41), which is the capillary path along which liquid sealant is drawn into the valve core. In this arrangement the clearance between the inner surface of the seat ring (30) and the poppet stem (41) at that portion is closed, or is sufficiently small to carry no significant flow, in the closed state and the normal-open state, so that the purge gap (60) remains a flow path distinct from the main passage (64).

[0128] In a fourth arrangement the purge gap (60) is formed by one or more openings through the wall of the deflectable ring element. The openings may be slits or apertures cut through the wall of the seat ring (30), the axial slots between the fingers of a slotted wall (332), or the cells of a cellular wall (432), as described with reference to FIG. 12. The openings are closed in the closed state and the normal-open state, either because the strained wall holds them shut or because they lie against the core body bore (22), and they are opened or uncovered when the purge shoulder (44) strains the wall during over-travel. The openings are placed where the wall moves away from the surface that closes them.

[0129] Accordingly, the purge opening through which the air purge passes may be an annular gap between a surface of the deflectable ring element and an adjacent surface of the core body (20), the valve stem (10) or the poppet (40); one or more slits or apertures through the wall of the deflectable ring element; or one or more gaps between the fingers of a segmented deflectable ring element. The deflectable ring element may correspondingly be continuous around its circumference, segmented by slots that leave a continuous band at the sealing edge (34), or split by a single axial split. A split deflectable ring element, in which the split itself is the purge opening and opens circumferentially under a tapered purge shoulder (244), is used only as a wiper ring (132) together with a fixed seat (130), because a split through a seat ring (30) would interrupt the sealing edge (34).

Alternative purge flow paths

[0130] In the embodiment of FIG. 3 the purge flow path (62) runs from the supply side, through the purge gap (60), across the surfaces of the valve core on which sealant film accumulates, and into the inflatable chamber alongside the main passage (64). The purge flow path (62) may take any of the following alternative forms, in each of which it remains distinct from the main passage (64) and carries inflation gas in the inflation direction into the inflatable chamber.

[0131] As shown in FIG. 13A, the purge flow path (62) may include one or more purge ports (262) formed as radial holes or slots through the wall of the core body (20) on the chamber side of the purge gap (60). The purge ports (262) lead from the purge gap (60) to the clearance between the core body (20) and the stem bore (12), and thence into the inflatable chamber. The purge ports (262) fix the position at which the air purge, and the sealant film it carries, leave the valve core. Two or more purge ports (262) may be distributed around the circumference of the core body (20).

[0132] As shown in FIG. 13B, the purge flow path (62) may include one or more flutes (362) formed as axial grooves in the core body bore (22) at the axial position of the seat ring (30). In the closed state and the normal-open state the outer surface of the seat ring (30) lies on the lands between the flutes (362), closing them. When the bounding portion of the outer surface of the seat ring (30) moves away from the core body bore (22) in the purge state, the flutes (362) are placed in communication with the supply side and carry the air purge past the seat ring (30). The flutes (362) set the flow area of the purge flow path (62) independently of the size of the purge gap opening, so that the flow area can be chosen by the number, width and depth of the flutes (362) rather than by the radial deflection alone.

[0133] Where the purge gap (60) is formed by openings through the wall of the deflectable ring element, as described above with reference to the slotted wall (332) and the cellular wall (432), the purge flow path (62) passes through the wall of the deflectable ring element and then along the poppet stem (41) or the core body bore (22) into the inflatable chamber. The purge ports (262) and the flutes (362) may be combined with any of the locations of the purge gap (60) described in the preceding section, and with one another.

Alternative deflecting formations

[0134] The purge shoulder (44) is one species of a deflecting formation, which is any formation that moves with the poppet (40) and that contacts the deflectable ring element only during over-travel, so as to deflect it radially and open the purge gap (60). In every deflecting formation described here the same poppet (40) performs both ranges of movement: during the normal stroke the deflecting formation is spaced from the deflectable ring element and the main passage (64) is open, and during over-travel the same poppet (40) carries the deflecting formation against the deflectable ring element. There is no second valve member and no second seat to be fouled by sealant film. In the embodiment of FIG. 1 the deflecting formation is an annular purge shoulder (44) integral with the poppet (40), on the chamber side of the sealing face (42), which bears axially on the seat ring (30) during over-travel so that the convoluted wall (32) converts the axial load into radial deflection.

[0135] As shown in FIG. 11A, the deflecting formation may be a tapered purge shoulder (244) having a frustoconical flank on the poppet (40) that enters the seat ring (30) during over-travel and deflects the seat ring (30) radially outward by direct radial contact, in the manner in which a mandrel expands a ring. Because the load applied by the tapered purge shoulder (244) is radial rather than axial, the seat ring (30) need not have an axial-to-radial conversion geometry, and a seat ring (30) with a plain cylindrical wall may be used with the tapered purge shoulder (244). Because outward expansion moves the outer surface of the seat ring (30) towards the core body bore (22), the tapered purge shoulder (244) is used with a purge gap (60) that outward deflection opens: a purge gap (60) at the inner surface of the seat ring (30) as in the third arrangement, an opening through the wall of the seat ring (30) as in the fourth arrangement, or the split of a split wiper ring (132) used with a fixed seat (130). The flank of the tapered purge shoulder (244) slides on the inner surface of the seat ring (30) during over-travel and during the return, and thereby also wipes that inner surface. The tapered purge shoulder (244) may equally be a taper on the poppet stem (41).

[0136] As shown in FIG. 11B, the deflecting formation may be provided as two or more discrete lugs (344) on the poppet (40), for example two to four discrete lugs (344) spaced around its circumference, in place of a continuous annular shoulder. The discrete lugs (344) load the deflectable ring element at discrete points, so that the ring ovalises, moving outward at the loaded points and inward between them, and the purge gap (60) opens over arcs of the circumference rather than all round. Less force is needed to deflect the deflectable ring element by discrete lugs (344) than by a continuous annular shoulder, and the purge flow is correspondingly non-uniform around the circumference.

[0137] As shown in FIG. 11C, the deflecting formation may be a separate collar (444) fixed on the poppet stem (41), for example pressed, crimped or threaded onto it. The separate collar (444) performs the function of the purge shoulder (44) and allows the material that contacts the deflectable ring element to be chosen independently of the material of the poppet (40); for example a separate collar (444) of stainless steel may be fitted to a poppet (40) of brass.

[0138] In further embodiments the deflecting formation is a return-spring retainer that seats the return spring (50) on the poppet (40) and that bears on the end of the seat ring (30) remote from the sealing edge (34) during over-travel, so that one part both retains the return spring (50) and deflects the seat ring (30); the load-receiving surface is then the end face of the seat ring (30) remote from the sealing edge (34). In other embodiments the deflecting formation is carried on a purge sleeve that slides on the poppet stem (41): the purge sleeve travels with the poppet (40) during the normal stroke and is driven onto the seat ring (30) only during over-travel, the sealing face (42) remaining on the poppet (40) itself. In other embodiments the deflecting formation is the sealing face (42) itself, or a continuation of the sealing face (42), which seals on the sealing edge (34) in the closed state and, when driven further during over-travel, forces the sealing edge (34) to deflect radially; the load-receiving surface is then the region of the sealing edge (34). In every one of these embodiments the deflecting formation is positioned so that it moves towards the load-receiving surface of the deflectable ring element as the poppet (40) moves in the opening direction, and is spaced from it in the closed state by the normal stroke.

[0139] The deflectable ring element and the deflecting formation may also be exchanged between the poppet (40) and the core body (20). In such an inverted arrangement the deflectable ring element is a thin skirt carried by the poppet (40), a fixed formation in the core body (20) deflects the skirt during over-travel to open the purge gap (60), and on shape recovery the skirt moves across the core body bore (22). The arrangement of FIG. 1 to FIG. 4, in which the deflectable ring element is held in the core body (20) and the deflecting formation moves with the

poppet (40), is preferred because the shape recovery of the held seat ring (30) then moves the seat ring (30) across the sealing face (42), which is the surface that seals and the surface on which sealant film matters, and because the wiping element is the seat itself, so that no separate wiping part is added that could itself be fouled.

Alternative threshold-setting arrangements

[0140] The second spring stage (54) of the return spring (50) is one species of a threshold-setting arrangement, which is any arrangement by which the force resisting movement of the poppet (40) rises at the end of the normal stroke, so that the pressure differential must exceed an over-travel threshold, higher than the crack pressure, before the poppet (40) enters over-travel. The over-travel threshold is what allows the valve core to inflate and to seal in the same way as a conventional valve core until the purge state is reached: below the over-travel threshold only the main passage (64) is open and the deflectable ring element is not deflected so as to open the purge gap (60). The deflecting formation may be spaced from the deflectable ring element throughout the normal stroke, as in the illustrated embodiment, or may contact the deflectable ring element at the end of the normal stroke and load it elastically, below the over-travel threshold, without opening the purge gap (60); in either case the valve core is in the normal-open state, which is defined by the main passage (64) being open and the purge gap (60) being closed. The threshold-setting arrangement may be provided in any of the following ways, and in each of them the over-travel threshold is set as a relationship to the crack pressure and to the pressure that the inflation device supplies, rather than by any one spring construction. In every species the over-travel threshold is verified on the assembled valve core by measuring the pressure differential at which the deflecting formation begins to deflect the deflectable ring element so as to open the purge gap (60).

[0141] In the embodiment of FIG. 1 the return spring (50) is a compression spring having a first spring stage (52) of lower rate, which resists the normal stroke and sets the crack pressure, and a second spring stage (54) of higher rate, which the pressure differential must overcome to produce over-travel and which sets the over-travel threshold. In one construction of such a return spring (50) the coils are of variable pitch, and the close-pitched section becomes solid at the end of the normal stroke, leaving fewer active coils and so a higher rate for the second spring stage (54).

[0142] As shown in FIG. 14A, the threshold-setting arrangement may comprise an auxiliary spring (254) in addition to the return spring (50). The return spring (50) then acts alone over the normal stroke as the first spring stage (52), and the auxiliary spring (254) is engaged only when a lost-motion clearance between the auxiliary spring (254) and the poppet (40) closes at the end of the normal stroke, after which the auxiliary spring (254) acts in parallel with the return spring (50). The two stages can then be tuned independently, and the lost-motion clearance is a feature that can be seen and measured on the assembled valve core. The auxiliary spring (254) may be a compression spring of wire, a stack of one or more conical disc springs, which gives a high force in a short axial length, or an annular compression bush of an engineering polymer.

[0143] As shown in FIG. 14B, the threshold-setting arrangement may comprise a detent (354) that holds the poppet (40) at the end of the normal stroke until the force on the poppet (40) exceeds a breakaway value. The detent (354) may be a spring-loaded ball, or a radially sprung element, carried in the core body (20) and engaging a circumferential groove on the poppet stem (41) at the end of the normal stroke; or it may be a permanent magnet carried in the core body (20) that attracts a ferromagnetic target, for example of ferritic stainless steel, carried on the poppet (40). After breakaway the resisting force falls, so that the poppet (40) passes through over-travel quickly, and the detent (354) releases on the return so that the poppet (40) can close. The magnetic form of the detent (354) has no sliding contact exposed to liquid sealant.

[0144] The threshold-setting arrangement may also be provided by the stiffness of the deflectable ring element itself, with a return spring (50) of a single rate. The over-travel threshold is then set by the force that the deflecting formation needs to deflect the deflectable ring element. In this arrangement the deflecting formation comes into contact with the deflectable ring element at the end of the normal stroke and, while the pressure differential is below the over-travel threshold, loads it elastically through the initial elastic rise of its loading curve without opening the purge gap (60); the valve core is then still in the normal-open state. Where the deflectable ring element is formed from a superelastic alloy, its loading follows a stress plateau after that initial elastic rise, so that the force needed to deflect it behaves as a threshold; the over-travel threshold set in this way depends on the dimensions of the deflectable ring element and on temperature. A deflectable ring element of a resilient polymer or of a resilient metal without a phase transformation has no such plateau, and the over-travel threshold set by its

stiffness alone is less distinct. In any of the arrangements described in this section the return spring (50) and the deflectable ring element may each contribute to the over-travel threshold.

[0145] In a further arrangement the deflecting formation is carried not by the poppet (40) but by a separate purge actuator, for example an annular piston or a flexible diaphragm housed in the core body (20), exposed on its supply side to the supply-side pressure and biased by its own compression spring. The purge actuator advances only when the pressure differential exceeds the over-travel threshold, carries the deflecting formation onto the deflectable ring element, and thereby opens the purge gap (60) while inflation gas is flowing, the poppet (40) performing only its normal stroke. The purge actuator adds a part and a sliding seal exposed to liquid sealant, and it is more readily accommodated in valve cores of larger bore, such as a Schrader-pattern valve core, than in a Presta-pattern valve core.

Alternative wiping arrangements

[0146] The snap-back wipe is the movement of the deflectable ring element across a surface of the poppet (40) as the deflectable ring element undergoes shape recovery when the pressure differential falls, by which movement partially cured sealant film is sheared from that surface. The snap-back wipe is defined by that movement and not by its speed or force. In the embodiment of FIG. 4 the recovering seat ring (30) moves across the sealing face (42), or across a surface of the poppet (40) adjacent to the sealing face (42). Shape recovery of a superelastic alloy is preferred because it is rapid and exerts a higher force, and so gives a stronger snap-back wipe; a seat ring (30) of a resilient polymer recovers more slowly and with less force, so that its snap-back wipe is weaker, but that movement is a snap-back wipe within the meaning used here, and the air purge provides a cleaning action independently of it.

[0147] In embodiments having a fixed seat (130) and a wiper ring (132), as shown in FIG. 10, the wiper ring (132) is deflected outward by the purge shoulder (44) during over-travel and undergoes shape recovery when the pressure differential falls. The wiper ring (132) may then perform the snap-back wipe on a surface of the poppet (40) in the same manner as the seat ring (30), while the fixed seat (130) performs the sealing function.

[0148] A mechanical wipe may additionally be provided by a structure that does not depend on shape recovery, in combination with the recovering deflectable ring element. In one such embodiment a fixed wiping land is formed as a close-clearance land in the core body bore (22), or on the fixed seat (130), through which the poppet (40) passes on every stroke, so that the relative axial movement between the poppet (40) and the fixed wiping land shears sealant film from the poppet (40); the deflection of the deflectable ring element then opens the purge gap (60) and the fixed wiping land supplements the snap-back wipe. In another embodiment a rotational wipe is provided by a helical guide between the poppet stem (41) and the core body (20), which turns the poppet (40) through a small angle during over-travel and back again on the return, so that the sealing face (42) rubs circumferentially across the sealing edge (34) in addition to the snap-back wipe.

[0149] The seat ring (30) may be provided with a defined wiping edge at its inner surface, being a sharp-cornered edge set by the cut profile of the seat ring (30) and preserved by limiting the extent of electropolishing, which is presented to the poppet (40) as the seat ring (30) recovers. The defined wiping edge concentrates the contact pressure so that the recovering seat ring (30) shears the sealant film rather than riding over it. The cleaning action of the valve core may be provided by the air purge, by the snap-back wipe, or by both, as described in relation to the operation of the valve core.

Alternative sealing faces

[0150] In the embodiment of FIG. 1 the sealing face (42) is a conical surface of the poppet (40) of brass, which seats on the sealing edge (34) of the seat ring (30) with metal-to-metal contact and without an elastomer. The sealing face (42) is one species of a sealing surface of the poppet (40), which may take any of the following alternative forms. The sealing face (42) may be part-spherical, giving line contact on the sealing edge (34) that tolerates misalignment of the poppet (40). The sealing face (42) may be a flat surface that bears on the axial end of the seat ring (30); such a flat sealing face (42) is less self-centring, and sealant film on the sealing surfaces has a greater effect on it.

[0151] The sealing face (42) may be of hard-anodised aluminium, the poppet (40) or a facing of the poppet (40) being of aluminium with a hard-anodised surface; a hard-anodised aluminium sealing face (42), or another wear-resistant finish of the sealing face (42), may be used where wear of the sealing face (42) from repeated metal-to-metal contact with the sealing edge (34) over the service life of the valve core is to be reduced. The sealing face (42) may be a polymer facing on the poppet (40), for example of PEEK, bearing on a metal sealing edge (34). The sealing face (42) may carry a hard low-adhesion coating, for example of electroless nickel or diamond-like carbon. A low-adhesion coating used alone only slows the build-up of sealant film and does not remove a film once formed; in the valve core described here such a coating is used together with the air purge and the snap-back wipe, which remove the film, so that the coating reduces the adhesion of what the cleaning action has to remove.

Materials of the core body

[0152] In the embodiment of FIG. 1 the core body (20) and the valve stem (10) are turned from free-machining brass CW614N, and the poppet (40) is of brass. The core body (20) may instead be formed from a corrosion-resistant material, and in particular from stainless steel, for example stainless steel 303, for corrosion resistance where the liquid sealant carried in the inflatable chamber contains ammonia. The choice of a stainless steel core body (20) is independent of the material and form of the deflectable ring element: a stainless steel core body (20) may be used with a seat ring (30) of a superelastic alloy, with a seat ring (30) of a resilient polymer, or with a wiper ring (132) used together with a fixed seat (130). The valve stem (10) and the poppet (40) may likewise be of stainless steel.

[0153] The core body (20) may also be formed from an engineering polymer, for example by injection moulding, in place of a metal. Where the core body (20) is of a resilient polymer or of spring-temper stainless steel, the deflectable ring element may be formed integrally with the core body (20) as a thin-walled flexure section, as described in relation to the deflectable ring element; a core body (20) of brass CW614N is not suitable for an integral deflectable ring element because the elastic strain that brass can recover is too small for the radial deflection required.

Retention of the deflectable ring element

[0154] In the embodiment of FIG. 1 and FIG. 8C the seat ring (30) is retained in the core body (20) by an interference fit of approximately 0.03 mm on diameter, and is seated axially against the internal shoulder (24). The interference fit is one species of ring retention, which is any means by which the deflectable ring element is held in the core body (20), or in the valve stem (10), against the internal shoulder (24) or an equivalent locating formation, so that the position of the sealing edge (34) is fixed while the portion of the deflectable ring element that undergoes radial deflection remains free to deflect. Whichever ring retention is used, it leaves that portion free.

[0155] The interference fit may be confined to an axial band of the seat ring (30) at the internal shoulder (24), the remainder of the outer surface of the seat ring (30) being free of the core body bore (22), as described with reference to FIG. 6. As shown in FIG. 8D, the seat ring (30) may instead be retained by a crimped lip (224) formed on the core body (20) and rolled or swaged over the end of the seat ring (30) remote from the internal shoulder (24); the crimped lip (224) retains the seat ring (30) axially and frees its outer surface from radial interference. As shown in FIG. 8E, the seat ring (30) may be retained by a retaining sleeve (324) pressed or threaded into the core body (20) and clamping the seat ring (30) against the internal shoulder (24), with the same freeing of the outer surface at the cost of one further part.

[0156] The deflectable ring element may also be retained by an adhesive bond, using an adhesive resistant to the liquid sealant and in particular to ammonia; by staking of the core body (20) over the deflectable ring element; by laser welding of the deflectable ring element to the core body (20); by over-moulding of a polymer core body (20) around the deflectable ring element; or by clamping of the deflectable ring element between two parts of the core body (20). The deflectable ring element may be formed integrally with the core body (20), in which case no separate ring retention is needed, or may be carried by the valve stem (10) rather than by the core body (20), as described in relation to the deflectable ring element and in the section on valve formats below.

[0157] Any of the ring retention means described here may be combined with one another. In some embodiments an interference fit is combined with a supplementary retention feature, for example an adhesive bond, a crimped lip (224) or a groove in the core body bore (22) into which the deflectable ring element snaps, so that the deflectable ring element cannot rotate or migrate axially in the core body bore (22), and cannot come free into the inflatable chamber, if the interference fit relaxes with repeated thermal cycling or vibration.

Valve formats and systems

[0158] The valve core is described above as a Presta-pattern valve core, whose core body (20) is threaded into the stem bore (12) of a Presta-pattern valve stem (10) of 6 mm outer diameter. The Presta-pattern valve core is the worked example and not a limitation: the operation of the deflectable ring element, the deflecting formation, the threshold-setting arrangement and the purge gap (60) does not depend on the thread by which the core body (20) is received, on the outer diameter of the valve stem (10), or on the Presta pattern. The same mechanism may be built into a valve core of another pattern, and into an inflation valve that has no removable valve core.

[0159] As shown schematically in FIG. 15A, the mechanism may be provided in a Schrader-pattern valve core, having a Schrader-pattern core body (620) and a Schrader-pattern poppet (630) which carry the seat ring (30), the purge shoulder (44) and the purge gap (60) in the same relationship as in FIG. 1 to FIG. 4. The larger bore of a Schrader-pattern valve core provides more room for the components of the mechanism, and in particular for a separate purge actuator as described above; and in a Schrader-pattern valve core the Schrader-pattern poppet (630) may be depressed by the head of the inflation device, which is one of the ways of producing over-travel described in relation to the operation of the valve core. The mechanism may likewise be provided in a Dunlop-pattern valve core.

[0160] As shown schematically in FIG. 15B, the mechanism may be provided in a non-removable inflation valve, in which the valve stem (10) directly carries a seat ring (730), together with the poppet (40) and the return spring (50), and there is no separate core body (20). The valve stem (10) then performs the functions that the core body (20) performs in the removable valve core, providing the locating formation against which the seat ring (730) is retained and the bore in which the poppet (40) is guided. Any of the ring retention means described above may be used to retain the seat ring (730) in the valve stem (10). The non-removable inflation valve is suited in particular to inflation valves fitted as original equipment to tubeless bicycle tyre set-ups.

[0161] An inflation valve according to this description comprises the valve stem (10) and, where the valve core is removable, the valve core received in the stem bore (12). The inflation valve may be fitted to a bicycle inner tube or to a tubeless bicycle tyre set-up, and the inflatable chamber so formed, holding liquid sealant, together with the inflation valve, forms a further aspect of this description. So does a wheel comprising a tubeless bicycle tyre set-up, or a bicycle inner tube, defining the inflatable chamber and the inflation valve mounted to the wheel in communication with the inflatable chamber, and so does a bicycle incorporating such a wheel. The mechanism is described with reference to inflation valves for bicycle inflatable chambers that hold liquid sealant, which is the application that has been built and tested; the same mechanism may be applied to inflation valves for other inflatable chambers that hold liquid sealant, and to other fluid valves in which a deposit forms on the sealing surfaces and is removed by a purge flow and by the recovery of a deflectable ring element.

Component and kit supply

[0162] The valve core is a unit that is separable from the valve stem (10), and it may be supplied as a replacement valve core for fitting into the stem bore (12) of an existing valve stem (10) in place of a conventional valve core, so that an inflatable chamber already fitted with a conventional inflation valve is given the cleaning action described here without replacement of the valve stem (10). The replacement valve core may be supplied singly or in a multipack, and may have a seat ring (30) of a superelastic alloy, a seat ring (30) of a resilient polymer, or a fixed seat (130) with a wiper ring (132).

[0163] The seat ring (30) may also be supplied as a component on its own, for fitting into a core body (20). As a component the seat ring (30) is adapted to the valve core described herein by the features stated in relation to the parameter relationships: a continuous sealing edge (34) at its first axial end on which the sealing face (42) of a poppet (40) is seatable; a load-receiving surface spaced axially from the sealing edge (34) on which a purge shoulder (44) can bear; a convoluted wall (32) between them that converts the axial load so applied into a radial

deflection at least equal to the purge gap opening within the strain the material recovers; an outer surface which, with the core body bore (22), bounds the purge gap (60); and shape recovery to its at-rest form on removal of the axial load. The seat ring (30) may further be supplied as a kit together with a poppet (40) having a sealing face (42) and a purge shoulder (44), and a return spring (50) having a first spring stage (52) and a second spring stage (54), for fitting into a core body (20) supplied separately, as indicated in FIG. 5. In such a kit the seat ring (30), the poppet (40) and the return spring (50) are matched so that, when assembled into the core body (20), the purge shoulder (44) is spaced from the seat ring (30) throughout the normal stroke and bears on the seat ring (30) only in over-travel, and the purge gap (60) is closed in the closed state and the normal-open state.

Thermally actuated purge

[0164] As a separate concept, the purge gap (60) may be opened by warming the deflectable ring element rather than by over-travel. In such an embodiment the deflectable ring element is formed from a two-way shape-memory alloy of nickel and titanium having an austenite finish temperature of about 35 °C to 40 °C, so that the deflectable ring element changes shape and opens the purge gap (60) when it is warmed above that temperature, for example by hand, and closes the purge gap (60) when it cools again. This thermally actuated purge allows the purge gap (60) to be opened without an inflation device, for a manual purge, and is described here as a concept distinct from the over-travel-actuated purge state of the embodiments described above; it has not been built or tested.

Combinations of the features described

[0165] The features described in this section and in the preceding sections on the deflectable ring element may be combined with one another and with each of the three principal embodiments described in the examples, namely the embodiment in which the deflectable ring element is a seat ring (30) formed from a superelastic alloy, the embodiment in which the deflectable ring element is a seat ring (30) formed from a resilient polymer, and the embodiment in which the deflectable ring element is a wiper ring (132) used together with a fixed seat (130). Each of the alternative locations of the purge gap (60), the alternative purge flow paths (62), the alternative deflecting formations, the alternative threshold-setting arrangements, the alternative wiping arrangements, the alternative sealing faces (42), the alternative materials of the core body (20), the alternative ring retention means and the alternative valve formats described above is disclosed in combination with each of the others and with each of the three principal embodiments, except where this description states an incompatibility or a dependency.

[0166] The stated incompatibilities and dependencies are these. Electropolishing is a finishing process for a metal deflectable ring element and does not apply to a deflectable ring element of a resilient polymer. An austenite finish temperature is a property of a superelastic alloy: where the deflectable ring element is formed from a superelastic alloy, the alloy is selected to have an austenite finish temperature below a threshold, such as 5 °C, so that the deflectable ring element remains superelastic across the intended range of ambient temperatures; no austenite finish temperature applies to a deflectable ring element of a resilient polymer, or to a wiper ring (132) of a resilient metal that is not a superelastic alloy. A split deflectable ring element is used only as a wiper ring (132) with a fixed seat (130). A seat ring (30) with a plain cylindrical wall is used with a tapered purge shoulder (244) or another deflecting formation that applies a radial load. An integral deflectable ring element requires a core body (20) of a resilient polymer or of spring-temper stainless steel. A separate purge actuator is more readily accommodated in a valve core of larger bore than the Presta-pattern valve core.

[0167] In some embodiments, therefore, the deflectable ring element is a seat ring (30) formed from a superelastic alloy, such as an alloy of nickel and titanium, having an austenite finish temperature below 5 °C; in other embodiments the deflectable ring element is a seat ring (30) formed from a resilient polymer, such as PEEK; and in other embodiments the deflectable ring element is a wiper ring (132) separate from a fixed seat (130). The core body (20) may be formed from a corrosion-resistant material, such as stainless steel, independently of which of these forms the deflectable ring element takes. The convoluted wall (32) described for converting axial load into radial deflection may be applied to a seat ring (30) that also performs the sealing function or, in embodiments having a separate fixed seat (130), may be applied to a wiper ring (132) that performs the purging and wiping function without also sealing. A lock nut (70) for holding the poppet (40) in the closed state by hand may be combined with any of these embodiments.

[0168] In embodiments in which the deflectable ring element is formed from a resilient polymer, such as PEEK, the shape recovery on release of the pressure differential may be slower, and the resulting snap-back wipe weaker, than in embodiments in which the deflectable ring element is formed from a superelastic alloy; the purge gap (60) and the air purge nonetheless provide a cleaning action independently of the snap-back wipe. The dimensions and pressure values given in the examples are those of the embodiment in which the seat ring (30) is formed from a superelastic alloy, the only embodiment built; they are applied to a seat ring (30) of a resilient polymer, to a wiper ring (132) used with a fixed seat (130), and to the alternative deflecting formations, threshold-setting arrangements and valve formats described in this section subject to the parameter relationships stated above, as described in relation to the dimensions and parameter ranges.

Manufacture of the seat ring

[0169] In the illustrated embodiment the seat ring (30) is made from tube stock of a superelastic alloy, NiTi of nominally 55.8 wt% nickel content, having an outer diameter of 2.7 mm and a wall of 0.20 mm. As shown in FIG. 8A, the tube stock is mounted on a mandrel and the profile of the seat ring is cut from it by laser cutting; the cut seat ring (30) before shape-setting is referred to herein as the seat ring blank. The convoluted wall (32) may be defined by the laser-cut profile, by forming on the mandrel, by the shape-setting fixture described below, or by a combination of these steps; the disclosure of the convoluted wall does not depend on which of these steps imparts the convolution, provided that the finished seat ring has the convoluted wall and the sealing edge (34) described above.

[0170] As shown in FIG. 8B, the seat ring blank is then subjected to shape-setting: it is held in a fixture that constrains the convoluted wall (32) to its intended wave geometry and is heated, in the illustrated embodiment at 500 °C for 10 minutes, and then water-quenched; the seat ring (30) after shape-setting is referred to herein as the shape-set seat ring. Shape-setting fixes the at-rest geometry of the seat ring, which is the geometry to which the seat ring returns by shape recovery when the pressure differential falls. Shape-setting time and temperature, together with the nickel content, also determine the transformation temperatures of the superelastic alloy. In the illustrated embodiment the seat ring is specified to have an austenite finish temperature below 5 °C, so that it is fully austenitic and behaves superelastically at ordinary riding temperatures. The values of 500 °C and 10 minutes are those used for the prototype seat rings of Example 1; in some embodiments the shape-setting temperature and time are selected so as to hold the austenite finish temperature of production seat rings below the specified value, and in some embodiments the austenite finish temperature is verified on sampled seat rings after shape-setting.

[0171] After shape-setting, the shape-set seat ring (30) is electropolished to remove the surface oxide and the heat-affected edge left by laser cutting. The seat ring wall thickness after electropolishing lies within the range stated for the seat ring wall thickness, and in some embodiments the seat ring wall thickness is checked after electropolishing, because electropolishing removes material from the whole surface including the root of each convolution of the convoluted wall (32). In other embodiments the seat ring is finished by mechanical or chemical polishing in place of electropolishing. In some embodiments the depth of electropolishing is limited at the inner surface of the seat ring so as to preserve a defined wiping edge, formed by the laser-cut profile, that is presented to the poppet (40) during the snap-back wipe; electropolishing rounds edges, and a limited depth of electropolishing retains the sharper edge while still removing the oxide.

[0172] Each step of this route determines a feature on which the operation of the valve core depends. Laser cutting from tube stock leaves the band of the seat ring (30) that carries the sealing edge (34) continuous, with no seam, so that the sealing face (42) seals on it around its whole circumference. Shape-setting in the fixture fixes the at-rest form of the seat ring (30), which is the form in which the purge gap (60) is closed and to which the seat ring (30) returns by shape recovery, and fixes the convolution amplitude which, with the seat ring wall thickness, sets the relationship between the axial load of the purge shoulder (44), the radial deflection and the strain in the wall; the shape-setting temperature and time, with the nickel content, set the austenite finish temperature on which superelastic recovery at riding temperature depends. Electropolishing sets the final seat ring wall thickness and, where its depth is limited at the inner surface, preserves the defined wiping edge. The route is therefore a route for making a seat ring (30) that seats a moving poppet (40), that is deflected radially by a mechanical load applied by the poppet (40) in over-travel, and that recovers on removal of that load at service temperature.

Alternative forming routes

[0173] The laser-cutting, shape-setting and electropolishing route is the route by which the seat rings (30) of Example 1 were made. The seat ring is not limited to that route, and any route that yields a deflectable ring element having the sealing edge (34), the convoluted wall (32) or another axial-to-radial conversion geometry, and the at-rest geometry described herein may be used. In some embodiments the convoluted wall of a seat ring of superelastic alloy is formed by roll-forming with profiled rollers, in stages, with shape-setting between the stages. In some embodiments the convoluted wall is formed by internal pressure in a split die; this route is more readily applied to a seat ring of stainless steel than to one of superelastic alloy at room temperature. In embodiments in which the seat ring is formed from a resilient polymer, such as PEEK, the seat ring may be injection moulded, and the shape-setting and electropolishing steps described above do not apply. In embodiments having a fixed seat (130) and a separate wiper ring (132), the wiper ring may be stamped or deep drawn from strip, as may a deflectable ring element in the form of a conical annular disc. A slotted wall (332) or a cellular wall (432) may be produced by laser cutting from tube stock as described above, or by photochemical etching of flat sheet followed by rolling and joining; the joined seam is a site of reduced fatigue life in a superelastic alloy, and for that material laser cutting from tube stock is preferred. None of the alternative forming routes of this section has been practised for the valve core described herein; they are described so that the seat ring is not limited to the route by which the prototype seat rings were made.

Assembly of the valve core

[0174] The core body (20) and the poppet (40) of the illustrated embodiment are turned from bar by ordinary machining, with no special process. As shown in FIG. 8C, the seat ring (30) is pressed into the core body bore (22) until it seats against the internal shoulder (24), with an interference fit of approximately 0.03 mm on diameter. The interference fit retains the seat ring axially against the internal shoulder and rotationally in the core body. As described above in connection with the closed state, in some embodiments the interference fit is confined to an axial band of the seat ring adjacent the internal shoulder, the remainder of the outer surface of the seat ring being free of the core body bore.

[0175] In other embodiments the seat ring (30) is retained in the core body (20) by ring retention other than, or in addition to, the interference fit. As shown in FIG. 8D, a crimped lip (224) of the core body may be rolled or swaged over the end of the seat ring remote from the internal shoulder (24). As shown in FIG. 8E, a retaining sleeve (324) pressed or threaded into the core body may clamp the seat ring against the internal shoulder. Each of these frees the outer surface of the seat ring from radial interference along its length. Further ring retention species, and their combination with the interference fit as supplementary retention, are described above in connection with the alternatives. In some embodiments the assembly step includes a check of the press-in force, or of the press-in travel, of the seat ring, so that a seat ring pressed with an interference at the tight end of the manufacturing tolerance is detected before it is assembled into a valve core in which the interference could suppress the radial deflection of the seat ring.

[0176] With the seat ring (30) retained, the poppet (40) is inserted so that its poppet stem (41) is guided by the core body (20) and its sealing face (42) rests on the sealing edge (34) of the seat ring, the return spring (50) is fitted so that its first spring stage (52) biases the poppet towards the closed state, and the lock nut (70) is fitted. The assembled valve core is threaded into the stem bore (12) of the valve stem (10). Where the valve core is supplied as a replacement valve core, this last step is performed by the user on an existing valve stem. Where the seat ring, or the seat ring with the poppet and the return spring, is supplied as a kit, the remaining assembly steps are performed by the recipient.

Examples

[0177] Example 1 and Comparative Example 1 report a bench test that was actually performed; every value in them is a measured or as-used value from that test. Examples 2, 3 and 4 are prophetic examples: they describe embodiments that have been conceived but not built or tested, they are written in the present or conditional tense, and no result is reported for them.

Example 1 (actual): superelastic seat ring embodiment

[0178] Three valve cores, designated P-1, P-2 and P-3, were built to the illustrated embodiment. Each comprised a core body (20) and valve stem (10) of free-machining brass CW614N, the valve stem being a Presta-pattern valve stem of 6 mm outer diameter; a seat ring (30) of superelastic alloy, NiTi of nominally 55.8 wt% nickel content, specified to have an austenite finish temperature below 5 °C, with a convoluted wall (32) and a sealing edge (34); a poppet (40) of brass with a conical sealing face (42) seating metal to metal on the sealing edge, and a purge shoulder (44) on the chamber side of the sealing face; a return spring (50) of stainless steel music wire having a first spring stage (52) and a second spring stage (54); a purge gap (60) closed in the closed state; and a lock nut (70).

[0179] The seat rings (30) of Example 1 had a seat ring inner diameter of 2.3 mm, a seat ring wall thickness of 0.20 mm, a seat ring free length of 3.0 mm and a convolution amplitude of 0.12 mm, and were made from tube stock of 2.7 mm outer diameter and 0.20 mm wall by the laser-cutting, shape-setting and electropolishing route described above, and were pressed into the core body (20) against the internal shoulder (24) with an interference fit of approximately 0.03 mm on diameter. The convoluted wall (32) of the seat rings of Example 1 had one convolution or two convolutions; the count of convolutions of the individual prototype seat rings is not recorded herein, and the worked example is not to be read as limited to either count. The valve cores of Example 1 had a crack pressure of 1.0 bar, a normal stroke of 0.6 mm, an over-travel threshold of 3.5 bar, an over-travel distance of 0.5 mm and a purge gap opening of 0.15 mm radial. The crack pressure, normal stroke, over-travel threshold, over-travel distance and purge gap opening are the design values of the prototype valve cores; the over-travel distance and the purge gap opening were not measured during the bench test.

[0180] The bench test was run from 12 August 2026 to 20 August 2026. Each valve core of Example 1, and each conventional valve core of Comparative Example 1, was fitted to an identical inflatable chamber, a 700×28 bicycle inner tube, charged with 60 ml of liquid sealant, a latex emulsion containing ammonia and suspended particles. Each inflation cycle consisted of an inflation event in which the inflatable chamber was inflated from fully deflated to 6 bar with the same inflation device, a floor pump, by the same operator; a hold of 24 hours at 22 °C; and full deflation through the inflation valve. The inflation time was measured from coupling of the inflation device to the supply side until the inflatable chamber reached 6 bar. Leak-down was measured over each 24-hour hold. A valve core was recorded as failed at the first inflation cycle at which its leak-down exceeded 0.5 bar in 24 hours, or at which the inflation device failed to open the valve within 5 seconds of engagement, the latter being recorded as blockage.

[0181] Each of the valve cores P-1, P-2 and P-3 completed 40 inflation cycles without failure, at which point the test was stopped. The inflation time of P-1 was 43 s at inflation cycle 1 and 47 s at inflation cycle 40; that of P-2 was 41 s at inflation cycle 1 and 46 s at inflation cycle 40; and that of P-3 was 44 s at inflation cycle 1 and 50 s at inflation cycle 40. The leak-down at inflation cycle 40 was 0.1 bar in 24 hours for P-1, 0.1 bar in 24 hours for P-2 and 0.2 bar in 24 hours for P-3. On every inflation event of every valve core of Example 1, an audible snap was heard on removal of the inflation device. The audible snap indicates that a shape recovery event of the seat ring (30) occurred on removal of the inflation device at these test conditions; it is not a measurement of the over-travel distance, of the purge gap opening, or of the extent of the snap-back wipe.

[0182] After inflation cycle 40 the valve cores P-1, P-2 and P-3 were removed and inspected under a stereo microscope. In each, the sealing edge (34) of the seat ring (30) and the sealing face (42) of the poppet (40) were clean of sealant film. Cured sealant film was present on the outer surface of the seat ring in the region of the purge gap (60). The bench test did not include fatigue testing of the seat ring beyond the 40 inflation cycles, did not include testing at temperatures other than 22 °C, did not include cold-shock testing such as would occur with a compressed-CO2 inflation device, and did not separate the contribution of the air purge from that of the snap-back wipe.

Comparative Example 1 (actual): conventional valve core

[0183] Three conventional valve cores, designated C-1, C-2 and C-3, all commercial Presta-pattern valve cores of one manufacturer's standard type and lacking a deflectable ring element and a purge gap, were tested under the protocol of Example 1 on identical inflatable chambers with the same liquid sealant. C-1 failed by blockage at inflation cycle 14, its inflation time having risen from 41 s at inflation cycle 1 to 68 s at inflation cycle 13. C-2 failed by leak-down of 0.7 bar in 24 hours at inflation cycle 12, its inflation time having risen from 40 s at inflation cycle 1 to 44 s at inflation cycle 11. C-3 failed by blockage at inflation cycle 18, its inflation time having risen from 42 s at

inflation cycle 1 to 75 s at inflation cycle 17. The conventional valve cores thus exhibited both of the failure modes described in the background and defined herein, leak-down and blockage, between inflation cycles 12 and 18.

Example 2 (prophetic): resilient polymer seat ring embodiment

[0184] In this prophetic example the valve core of Example 1 is built with a seat ring (30) formed from a resilient polymer, PEEK, in place of the superelastic alloy, the seat ring having the same convoluted wall (32) and sealing edge (34), and the core body (20), poppet (40) with sealing face (42) and purge shoulder (44), return spring (50), purge gap (60) and lock nut (70) being as in Example 1. The seat ring of this example would be made by a route suited to the polymer, such as injection moulding, without the shape-setting and electropolishing steps of Example 1. On over-travel the purge shoulder would deflect the seat ring radially and open the purge gap, so that the air purge would operate as in Example 1. On removal of the inflation device the shape recovery of the polymer seat ring would be slower, and its recovery force lower, than the superelastic recovery of the seat ring of Example 1, so that the snap-back wipe would be weaker; the air purge would provide cleaning action independently of the snap-back wipe. No dimensions specific to the polymer seat ring are given herein; the person skilled in the art would select them within the ranges stated above so that the parameter relationships stated in relation to those ranges hold for the stiffness and recoverable strain of the resilient polymer. No inflation cycle, leak-down or inflation time result exists for this example.

Example 3 (prophetic): fixed seat and wiper ring embodiment

[0185] In this prophetic example, shown in FIG. 10A and FIG. 10B, the valve core comprises a core body (20) and valve stem (10) as in Example 1, a rigid fixed seat (130) of brass on which the sealing face (42) of the poppet (40) seats in the closed state, and a separate wiper ring (132) of stainless spring steel positioned on the chamber side of the fixed seat. The poppet, return spring (50) and lock nut (70) are as in Example 1. At and above the crack pressure the poppet would lift its sealing face from the fixed seat by the normal stroke, with the purge gap (60) closed. Above the over-travel threshold the purge shoulder (44) would deflect the wiper ring radially, opening the purge gap (60) bounded by a surface of the wiper ring, and the air purge would flow along the purge flow path (62) into the inflatable chamber; where the wiper ring is deflected outward, the purge gap would lie at its inner surface or through its wall, and where the purge gap lies between the outer surface of the wiper ring and the core body bore (22), as shown in FIG. 10B, it would be opened by inward or non-uniform movement of the bounding portion of the wiper ring, as described above. On removal of the inflation device the wiper ring would undergo shape recovery and the purge gap would close. This embodiment separates the sealing function, performed by the fixed seat, from the purging function and any wiping function, performed by the wiper ring, at the cost of one more part and greater axial length than Example 1. No result exists for this example.

Example 4 (prophetic): corrosion-resistant core body and wear-resistant sealing face

[0186] In this prophetic example the valve core of Example 1, Example 2 or Example 3 is built with a core body (20) of stainless steel 303 in place of brass CW614N, for corrosion resistance in service with a liquid sealant containing ammonia. A core body of stainless steel 303 has been machined; the bench test of Example 1 was performed with core bodies of brass only, and no result exists for a valve core having a core body of stainless steel 303. Independently of the core body material, in this prophetic example the poppet (40) carries a sealing face (42) of hard-anodised aluminium in place of the brass sealing face of Example 1, as a wear-resistant sealing face for repeated metal-to-metal seating on the sealing edge (34) of the seat ring (30) or on the fixed seat (130). The hard-anodised aluminium sealing face has been considered but not built. No result exists for either variant.

Data and results

[0187] The results of Example 1 and Comparative Example 1 are summarised in the following table. In the table, "censored" means that the test was stopped at inflation cycle 40 without failure of the valve core, so that the number of inflation cycles to failure is not known and is greater than 40.

Valve core	Type	Inflation cycles completed without failure	Failure mode	Inflation time at inflation cycle 1 (s)	Inflation time at last inflation cycle (s)	Leak-down at last inflation cycle (bar in 24 h)
P-1	Example 1	40 (censored)	none	43	47 (inflation cycle 40)	0.1
P-2	Example 1	40 (censored)	none	41	46 (inflation cycle 40)	0.1
P-3	Example 1	40 (censored)	none	44	50 (inflation cycle 40)	0.2
C-1	Comparative Example 1	13	blockage at inflation cycle 14	41	68 (inflation cycle 13)	not the failure criterion
C-2	Comparative Example 1	11	leak-down of 0.7 bar in 24 h at inflation cycle 12	40	44 (inflation cycle 11)	0.7 (inflation cycle 12)
C-3	Comparative Example 1	17	blockage at inflation cycle 18	42	75 (inflation cycle 17)	not the failure criterion

[0188] The scope of the data is as follows. Three valve cores were tested in each group. The valve cores of Example 1 were run for 40 inflation cycles without failure and the test was then stopped, so the result is censored and the fatigue life of the seat ring (30), and its behaviour beyond 40 inflation cycles, are not established by the data. One liquid sealant, one inflation device, one operator, one inflatable chamber type, one hold temperature of 22 °C and one inflation pressure of 6 bar were used. The over-travel distance and the purge gap opening were not measured during the test; over-travel and shape recovery are indicated only by the audible snap on removal of the inflation device. No valve core of Example 2, Example 3 or Example 4 was tested.

[0189] Within that scope, the data show that the valve cores of Example 1 continued to seal, with a leak-down at inflation cycle 40 of 0.1 to 0.2 bar in 24 hours against a failure criterion of 0.5 bar in 24 hours, and continued to open, with an inflation time at inflation cycle 40 within 15 % of that at inflation cycle 1, over 40 sealant-loaded inflation cycles, whereas the conventional valve cores of Comparative Example 1 failed by blockage or leak-down between inflation cycles 12 and 18. No tool and no user action other than ordinary inflation was applied to any valve core of Example 1 during the test.

[0190] The post-test inspection of Example 1 shows which surfaces of the valve core were kept clear of sealant film. The sealing edge (34) of the seat ring (30) and the sealing face (42) of the poppet (40), which are the surfaces whose contact seals the valve core in the closed state and the surfaces across which the seat ring moves during the snap-back wipe, were clean. Sealant film remained on the outer surface of the seat ring in the region of the purge gap (60), a surface that neither seals in the closed state nor is wiped by the seat ring. The data do not separate the cleaning action of the air purge from that of the snap-back wipe, and no attribution between them is made herein; the cleaning action described herein is the air purge, the snap-back wipe, or both.

Robustness and failure modes

[0191] The following paragraphs describe conditions under which the performance of the valve core may be degraded, and the features described herein that address them. Where a feature has been tested it is so stated; where a feature has not been tested, it is described as a feature that may be used and no result is asserted for it.

[0192] Sealant film was found after 40 inflation cycles on the outer surface of the seat ring (30) in the region of the purge gap (60), a region that is closed, and therefore not purged, whenever the valve core is not in the purge state. Whether that film accumulates further over a service life longer than 40 inflation cycles, and whether it

affects the opening or closing of the purge gap, was not tested. In some embodiments the purge gap opening is selected towards the upper end of the range of 0.10 mm to 0.25 mm, so as to increase the clearance and the velocity of the air purge in that region. In some embodiments the surface finish applied to the seat ring, such as electropolishing, is also applied to the adjoining surface of the core body bore (22) so as to reduce the adhesion of sealant film to it. The snap-back wipe occurring on every shape recovery may itself prevent sealant film in the closed purge gap from hardening; this has not been tested beyond 40 inflation cycles.

[0193] The seat ring (30) is subjected to radial deflection and shape recovery on every purge state, and the strain is concentrated at the root of each convolution of the convoluted wall (32). The fatigue life of the seat ring beyond the 40 inflation cycles of Example 1 has not been tested. In some embodiments the convoluted wall has two convolutions rather than one, so that the strain of a given radial deflection is distributed between them; in some embodiments a fillet radius is provided at the root of each convolution. No fatigue result is asserted for either feature.

[0194] The seat ring (30) of the illustrated embodiment is specified to have an austenite finish temperature below 5 °C, so that it is superelastic across ordinary riding temperatures. Its martensite start temperature has not been measured, and its behaviour if chilled below the martensite start temperature, whether by cold weather or by the expansion of gas from a compressed-CO2 inflation device while the seat ring is in the purge state, has not been tested; superelastic shape recovery may be impaired below the martensite start temperature until the seat ring warms. In some embodiments the superelastic alloy is selected, by its nickel content or by its shape-setting, to have a lower austenite finish temperature, giving more margin against chilling. In other embodiments the deflectable ring element is formed from a resilient metal or a resilient polymer whose shape recovery does not depend on a phase transformation, as described above in connection with the materials of the deflectable ring element, or the shape recovery of the deflectable ring element is positively driven by a formation of the poppet (40) on its return, as also described above. No value of the martensite start temperature and no cold-inflation margin is asserted herein.

[0195] Retention of the seat ring (30) in the illustrated embodiment relies on the interference fit of approximately 0.03 mm on diameter against the internal shoulder (24). The working range of the interference has not been determined, and no test of retention under thermal cycling or vibration has been performed. In some embodiments, supplementary ring retention is provided in addition to the interference fit, such as the crimped lip (224), the retaining sleeve (324), an adhesive bond, staking or laser welding described herein, so that the seat ring cannot migrate axially, rotate or come free into the inflatable chamber. In some embodiments the interference is held within a specified tolerance band and the press-in force or travel is checked at assembly, as described above, so that an interference at the tight end of the tolerance does not suppress the radial deflection of the seat ring or crack it at assembly, and an interference at the loose end does not leave the seat ring unretained.

[0196] In the illustrated embodiment the seal in the closed state is formed by metal-to-metal contact between the brass sealing face (42) of the poppet (40) and the sealing edge (34) of the seat ring (30). The bench test of Example 1 ran for 40 inflation cycles; the wear of the sealing face and the sealing edge over a service life of several thousand inflation cycles has not been tested. In some embodiments the sealing face is a hard-anodised aluminium sealing face, or carries a wear-resistant finish, as described in Example 4 and in connection with the alternatives above.

[0197] If the pressure differential rises rapidly and well beyond the over-travel threshold, for example on inflation from a shop compressor, the poppet (40) is driven into over-travel with greater force than on inflation with a floor pump. In some embodiments over-travel is limited independently of the deflectable ring element, as described above in connection with over-travel, so that the radial deflection of the seat ring (30) cannot exceed the recoverable strain of its material whatever the pressure differential applied. The bench test of Example 1 applied one inflation event per 24-hour inflation cycle with a floor pump; rapid repeated inflation events in a short period, inflation from a compressor, and the first inflation event after prolonged storage of the inflatable chamber in a deflated state, in which the sealant film near the inflation valve may have cured more fully than between the daily inflation cycles of the test, were not tested.

[0198] The austenite finish temperature, the convolution amplitude, the seat ring wall thickness after electropolishing and the over-travel threshold set by the second spring stage (54) each depend on a manufacturing step for which Example 1 gives the values used for the prototype valve cores and not a process

window. In some embodiments production seat rings (30) are verified by sampling the austenite finish temperature after shape-setting, by measuring the convolution amplitude before shape-setting and the seat ring wall thickness after electropolishing, and by sorting or verifying the rate of the second spring stage at assembly, so that each production valve core lies within the ranges stated herein. These verification steps are described as manufacturing options and have not been practised for the prototype valve cores.

[0199] The liquid sealant of Example 1 contained ammonia. Liquid sealants containing ammonia can attack brass over time. A core body (20) of stainless steel 303 is described in Example 4 and in connection with the alternatives above as an option for applications where this is a concern. Whether attack occurs in service on the core body of brass CW614N or on the poppet (40) of the illustrated embodiment has not been tested. Liquid sealants that are free of ammonia, and liquid sealants of higher viscosity or with abrasive particulate fillers, were not tested, and the effect of the valve core with them is not asserted herein.

[0200] For storage and transport the lock nut (70) is used to hold the poppet (40) in the closed state, in which the purge shoulder (44) is clear of the seat ring (30) and the seat ring is at rest. A valve cap or accessory having an internal pin that depresses the poppet should not be used in a manner that holds the poppet in over-travel, since that would hold the seat ring in sustained radial deflection rather than the transient radial deflection of the purge state. Because the seat ring wall thickness is between 0.15 mm and 0.30 mm, a tool inserted into the valve core in the manner used to clear a conventional valve core may damage the convoluted wall (32); the valve core described herein is intended to be cleaned by the air purge and the snap-back wipe without tool insertion.

Industrial applicability and use

[0201] The inflation valve and the valve core described herein are for use with an inflatable chamber that holds liquid sealant, including a bicycle inner tube, as tested in Example 1, and a tubeless bicycle tyre set-up, in which the liquid sealant is likewise injected through the valve stem (10) and drawn towards the valve core on each inflation event. Use in a tubeless bicycle tyre set-up has not been tested. The valve core opens, closes and enters the purge state in response to the pressure differential alone; no tool and no user action beyond coupling and removing the inflation device is required for either cleaning action.

[0202] The purge state is reached when the pressure differential applied by the inflation device exceeds the over-travel threshold, which in Example 1 was 3.5 bar and was exceeded on every inflation event with a floor pump inflating the inflatable chamber from fully deflated to 6 bar. Whether a given inflation device reaches the over-travel threshold depends on the pressure differential it applies at the supply side: it is expected that floor pumps of the kind used in Example 1 do so, and hand pumps, compressors and compressed-CO2 inflation devices may or may not do so depending on the pressure differential they generate; only the floor pump of Example 1 has been tested. An inflation event in which the pressure differential does not exceed the over-travel threshold inflates the inflatable chamber through the main passage (64) in the normal-open state without entering the purge state, the purge gap (60) remaining closed; the air purge and the snap-back wipe occur on the inflation events in which the over-travel threshold is exceeded. Any inflation gas may serve as the air purge.

[0203] The valve core may be supplied as original equipment fitted to the valve stem (10) of an inflation valve, or as a replacement valve core for fitting to an existing valve stem in place of a conventional valve core, in which case it is threaded into the stem bore (12) in the same manner as the conventional valve core it replaces, and the lock nut (70) of the existing inflation valve may be retained. As shown in FIG. 5, the seat ring (30), or the seat ring together with the poppet (40) and the return spring (50), may also be supplied as a kit for assembly into a core body (20), as described above in connection with component supply. The inflation valve may be supplied fitted to a bicycle inner tube, a tubeless bicycle tyre set-up, a wheel, or a bicycle.

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Fig. 1 — Axial section through the inflation valve of the illustrated embodiment, on a plane containing the poppet axis, in the closed state: valve stem (10), stem bore (12), core body (20), core body bore (22), internal shoulder (24), seat ring (30), convoluted wall (32), sealing edge (34), poppet (40), poppet stem (41), sealing face (42), purge shoulder (44), return spring (50), first spring stage (52), second spring stage (54), purge gap (60) closed, main passage (64) closed, lock nut (70). Fig. 2 — The section of Fig. 1 in the normal-open state: poppet (40) displaced through the normal stroke, sealing face (42) lifted from the sealing edge (34), main passage (64) open,

purge shoulder (44) spaced from the seat ring (30), purge gap (60) closed. Fig. 3 — The section of Fig. 1 in the purge state: poppet (40) in over-travel, purge shoulder (44) bearing on the seat ring (30), seat ring (30) in radial deflection with its at-rest outline in dashed line, purge gap (60) open, main passage (64) and purge flow path (62) shown with arrows. Fig. 4 — The section of Fig. 1 during shape recovery of the seat ring (30) after removal of the inflation device, showing the snap-back wipe across the sealing face (42) and the purge gap (60) closing. Fig. 5 — The components of the inflation valve separated along the poppet axis in assembly order: lock nut (70), valve stem (10), core body (20), seat ring (30), poppet (40) with sealing face (42) and purge shoulder (44), return spring (50) with first spring stage (52) and second spring stage (54). Fig. 6 — Enlarged details of the region of the seat ring (30), the internal shoulder (24), the core body bore (22) and the purge shoulder (44): Fig. 6A in the closed state with the convoluted wall (32), its convolutions (32a, 32b), the sealing edge (34), the interference-fit region and the seat ring dimensions called out; Fig. 6B in the purge state with the purge gap (60) and its purge gap opening called out. Fig. 7 — Diagram of the states of the valve core (closed, normal-open, purge, recovery and locked) and of the conditions governing the transitions between them. Fig. 8 — Manufacture and assembly of the seat ring (30): Fig. 8A laser cutting from tube stock on a mandrel; Fig. 8B shape-setting in a fixture; Fig. 8C pressing into the core body (20) against the internal shoulder (24); Fig. 8D retention by a crimped lip (224); Fig. 8E retention by a retaining sleeve (324). Fig. 9 — Section corresponding to Fig. 1 of an embodiment in which the seat ring (30) is formed from a resilient polymer. Fig. 10 — Sections of an embodiment having a fixed seat (130) and a separate wiper ring (132): Fig. 10A in the closed state; Fig. 10B in the purge state with the purge shoulder (44) deflecting the wiper ring (132) and the purge gap (60) open. Fig. 11 — Alternative deflecting formations of the poppet (40), each shown against the seat ring (30) in over-travel: Fig. 11A a tapered purge shoulder (244); Fig. 11B discrete lugs (344); Fig. 11C a separate collar (444) fixed on the poppet stem (41). Fig. 12 — Alternative wall geometries of the deflectable ring element replacing the convoluted wall (32): Fig. 12A a frustoconical wall (232); Fig. 12B a slotted wall (332); Fig. 12C a cellular wall (432). Fig. 13 — Alternative purge flow paths: Fig. 13A through purge ports (262) in the wall of the core body (20) on the chamber side of the purge gap (60); Fig. 13B along flutes (362) in the core body bore (22). Fig. 14 — Alternative threshold-setting arrangements replacing the second spring stage (54): Fig. 14A an auxiliary spring (254) engaged after a lost-motion clearance; Fig. 14B a detent (354). Fig. 15 — Schematic views of the mechanism in other valve architectures: Fig. 15A a Schrader-pattern valve core with a Schrader-pattern core body (620) and a Schrader-pattern poppet (630); Fig. 15B a non-removable inflation valve in which the seat ring (730), the poppet and the return spring are carried directly by the valve stem with no separate core body.

Claims

- A valve core for an inflation valve, the valve core comprising a core body (20), a poppet (40) guided by the core body, a deflectable ring element carried by the core body, a return spring (50) biasing the poppet towards a closed state of the valve core, and a main passage (64) past a sealing face (42) of the poppet, the main passage being closed by the sealing face in the closed state; the poppet having a supply side and a chamber side and being movable from the closed state in an opening direction through a normal stroke and beyond the normal stroke into over-travel; the valve core having: the closed state, in which the main passage is closed, the deflectable ring element is in its at-rest form and a purge gap (60) associated with the deflectable ring element is closed; a normal-open state, in which the poppet is displaced from the closed state within the normal stroke, the main passage is open and the purge gap is closed; and a purge state, in which the poppet is in over-travel beyond the normal stroke and produces a radial deflection of the deflectable ring element, the radial deflection opening the purge gap so that a purge flow path (62), separate from the main passage, extends from the supply side through the purge gap to the chamber side; and wherein, when the poppet returns from over-travel, the deflectable ring element undergoes shape recovery to its at-rest form and closes the purge gap.
- The valve core of claim 1, wherein the poppet is displaced from the closed state in response to a pressure differential between the supply side and the chamber side, the valve core being in the closed state at a pressure differential below a crack pressure, in the normal-open state at a pressure differential at or above the crack pressure and not exceeding an over-travel threshold higher than the crack pressure, and in the purge state at a pressure differential exceeding the over-travel threshold.
- The valve core of claim 1 or claim 2, wherein the poppet has a deflecting formation (44) which is spaced from the deflectable ring element in the closed state and throughout the normal stroke and which bears on the deflectable ring element in over-travel so as to produce the radial deflection.

- The valve core of claim 3, wherein the deflecting formation is a purge shoulder (44) of the poppet on the chamber side of the sealing face.
- The valve core of claim 4, wherein the purge shoulder is an annular shoulder integral with the poppet which bears axially on the deflectable ring element in over-travel.
- The valve core of claim 3, wherein the deflecting formation is a tapered purge shoulder (244) having a frustoconical flank which enters the deflectable ring element in over-travel and deflects the deflectable ring element radially outward by direct radial contact with an inner surface of the deflectable ring element.
- The valve core of claim 3, wherein the deflecting formation comprises two or more discrete lugs (344) spaced around a circumference of the poppet, which load the deflectable ring element at discrete points so that the purge gap opens over arcs of a circumference of the deflectable ring element.
- The valve core of claim 3, wherein the deflecting formation is provided on a separate collar (444) fixed on a poppet stem (41) of the poppet.
- The valve core of any one of claims 1 to 8, wherein the deflectable ring element is a seat ring (30) having a sealing edge (34) on which the sealing face rests in the closed state.
- The valve core of any one of claims 1 to 8, further comprising a fixed seat (130) carried by the core body, the fixed seat not deflecting in over-travel and providing a sealing edge on which the sealing face rests in the closed state, wherein the deflectable ring element is a wiper ring (132) separate from the fixed seat and positioned on the chamber side of the fixed seat.
- The valve core of any one of claims 1 to 10, wherein the deflectable ring element has a wall arranged such that an axial load applied to the deflectable ring element by the poppet in over-travel is converted into the radial deflection.
- The valve core of claim 11, wherein the wall comprises a convoluted wall (32) having one or more convolutions.
- The valve core of claim 12, wherein the convoluted wall has one convolution or two convolutions.
- The valve core of claim 12 or claim 13, wherein the convoluted wall has a convolution amplitude of from 0.08 mm to 0.20 mm.
- The valve core of any one of claims 12 to 14, wherein a root of each convolution of the convoluted wall is provided with a fillet radius.
- The valve core of claim 11, wherein the wall is a frustoconical wall (232), a slotted wall (332) or a cellular wall (432).
- The valve core of claim 9, wherein the seat ring has a seat ring inner diameter of from 2.0 mm to 2.6 mm and a seat ring wall thickness of from 0.15 mm to 0.30 mm.
- The valve core of claim 17, wherein the seat ring has a seat ring free length of from 2.5 mm to 4.0 mm.
- The valve core of any one of claims 1 to 18, wherein the core body has an internal shoulder (24) and the deflectable ring element is retained in the core body against the internal shoulder, the radial deflection being an elastic deformation of a wall of the deflectable ring element while the deflectable ring element remains retained in the core body.
- The valve core of claim 19, wherein the deflectable ring element is retained in a core body bore (22) of the core body by an interference fit.
- The valve core of claim 20, wherein the interference fit has an interference of approximately 0.03 mm on diameter.
- The valve core of claim 20 or claim 21, wherein the interference fit is provided over an axial band of the deflectable ring element adjacent the internal shoulder, and a remainder of an outer surface of the deflectable ring element, including a portion of the wall which undergoes the radial deflection, is not in interference with the core body bore.
- The valve core of any one of claims 19 to 22, wherein the deflectable ring element is retained by a crimped lip (224) formed on the core body over an end of the deflectable ring element remote from the internal shoulder,

or by a retaining sleeve (324) pressed or threaded into the core body and clamping the deflectable ring element against the internal shoulder.

- The valve core of any one of claims 19 to 23, wherein the deflectable ring element is further retained by an adhesive bond, by staking of the core body over the deflectable ring element, by laser welding of the deflectable ring element to the core body, by over-moulding of the core body around the deflectable ring element, or by clamping of the deflectable ring element between two parts of the core body.
- The valve core of any one of claims 1 to 18, wherein the deflectable ring element is an integral flexure section of the core body formed in one piece with the core body, the core body being formed from a resilient polymer or from stainless steel of spring temper.
- The valve core of any one of claims 1 to 25, wherein, as the deflectable ring element undergoes shape recovery, the deflectable ring element moves across a surface of the poppet and shears sealant film from that surface.
- The valve core of claim 26, wherein the surface of the poppet is the sealing face.
- The valve core of claim 26 or claim 27, wherein the deflectable ring element has a defined wiping edge at an inner surface of the deflectable ring element which is presented to the poppet as the deflectable ring element undergoes shape recovery.
- The valve core of any one of claims 1 to 28, wherein the poppet has a second formation, on a side of the deflectable ring element opposite to a side which the poppet contacts to produce the radial deflection in over-travel, which meets the deflectable ring element on a return movement of the poppet and drives the deflectable ring element towards its at-rest form.
- The valve core of any one of claims 1 to 29, wherein the radial deflection is radially outward.
- The valve core of any one of claims 1 to 29, wherein the radial deflection is radially inward at a surface of the deflectable ring element which bounds the purge gap.
- The valve core of claim 9, wherein the purge gap is an annular radial clearance between an outer surface of the seat ring and a core body bore (22) of the core body, which opens where a bounding portion of the outer surface moves away from the core body bore in the purge state.
- The valve core of claim 9, wherein the purge gap is bounded by an inner surface of the seat ring and the poppet on the chamber side of the sealing edge, and the purge flow path passes along a poppet stem (41) of the poppet.
- The valve core of any one of claims 1 to 31, wherein the purge gap is formed by one or more openings through a wall of the deflectable ring element, the openings being closed in the closed state and in the normal-open state and being opened or uncovered by the radial deflection in the purge state.
- The valve core of any one of claims 1 to 34, wherein the purge flow path includes one or more purge ports (262) formed through a wall of the core body on the chamber side of the purge gap.
- The valve core of any one of claims 1 to 35, wherein the purge flow path includes one or more flutes (362) formed as axial grooves in a core body bore (22) of the core body at an axial position of the deflectable ring element, an outer surface of the deflectable ring element lying on lands between the flutes and closing the flutes in the closed state and in the normal-open state.
- The valve core of any one of claims 1 to 36, wherein the purge gap has a purge gap opening of from 0.10 mm to 0.25 mm in the purge state.
- The valve core of claim 2, or of any one of claims 3 to 37 when dependent on claim 2, further comprising a threshold-setting arrangement by which a force resisting movement of the poppet rises at an end of the normal stroke, the threshold-setting arrangement setting the over-travel threshold.
- The valve core of claim 38, wherein the threshold-setting arrangement is a second spring stage (54) of the return spring, the return spring having a first spring stage (52) of lower rate which resists the normal stroke and sets the crack pressure, and the second spring stage having a higher rate than the first spring stage, coming into action when the normal stroke is complete and setting the over-travel threshold.

- The valve core of claim 38, wherein the threshold-setting arrangement comprises an auxiliary spring (254) which is engaged when a lost-motion clearance between the auxiliary spring and the poppet closes at the end of the normal stroke, after which the auxiliary spring acts in parallel with the return spring.
- The valve core of claim 38, wherein the threshold-setting arrangement comprises a detent (354) which holds the poppet at the end of the normal stroke until a force on the poppet exceeds a breakaway value.
- The valve core of claim 2, or of any one of claims 3 to 37 when dependent on claim 2, wherein the return spring has a single rate and the over-travel threshold is set by a stiffness of the deflectable ring element.
- The valve core of claim 2, or of any one of claims 3 to 42 when dependent on claim 2, wherein the over-travel threshold is from 2.5 bar to 5.0 bar.
- The valve core of claim 2, or of any one of claims 3 to 43 when dependent on claim 2, wherein the crack pressure is from 0.8 bar to 1.5 bar.
- The valve core of any one of claims 1 to 44, wherein in over-travel the poppet moves beyond an end of the normal stroke by an over-travel distance of from 0.4 mm to 0.9 mm.
- The valve core of claim 2, or of any one of claims 3 to 42 when dependent on claim 2, wherein the crack pressure is about 1.0 bar and the over-travel threshold is about 3.5 bar.
- The valve core of any one of claims 1 to 46, wherein the core body has an over-travel stop which the poppet meets at an end of over-travel, the over-travel stop limiting an over-travel distance of the poppet.
- The valve core of any one of claims 1 to 47, wherein the poppet is movable into over-travel by mechanical depression of the poppet by a depressor in a head of an inflation device, or in an adaptor fitted between the inflation device and the supply side, when the inflation device is coupled to the supply side.
- The valve core of any one of claims 1 to 48, wherein the purge flow path is arranged to carry flow through the purge gap in either direction between the supply side and the chamber side.
- The valve core of any one of claims 1 to 49, wherein the deflectable ring element is formed from a superelastic alloy.
- The valve core of claim 50, wherein the superelastic alloy has an austenite finish temperature below 5 °C.
- The valve core of claim 50 or claim 51, wherein the superelastic alloy is a nickel-titanium alloy of nominally 55.8 wt% nickel content.
- The valve core of claim 50 or claim 51, wherein the superelastic alloy is a NiTiCu alloy, a NiTiFe alloy or a NiTiCr alloy.
- The valve core of any one of claims 1 to 49, wherein the deflectable ring element is formed from a resilient metal which recovers its shape elastically without a phase transformation, the resilient metal being stainless steel of spring temper, cold-worked austenitic stainless steel, precipitation-hardened stainless steel or a beta-titanium alloy.
- The valve core of any one of claims 1 to 49, wherein the deflectable ring element is formed from a resilient polymer.
- The valve core of claim 55, wherein the resilient polymer is polyether ether ketone.
- The valve core of any one of claims 1 to 56, wherein the core body is formed from stainless steel.
- The valve core of any one of claims 1 to 57, wherein the sealing face is of hard-anodised aluminium or carries a hard low-adhesion coating.
- The valve core of any one of claims 1 to 58, wherein the core body has an external thread for engagement in a stem bore of a valve stem, the valve core being removable from the valve stem as a unit.
- The valve core of any one of claims 1 to 59, wherein the core body is a Schrader-pattern core body (620) and the poppet is a Schrader-pattern poppet (630).
- A valve core for an inflation valve, the valve core comprising a core body (20), a poppet (40) guided by the core body, a deflectable ring element carried by the core body, a return spring (50) biasing the poppet towards a closed state of the valve core, and a main passage (64) past a sealing face (42) of the poppet, the main

passage being closed by the sealing face in the closed state; the poppet having a supply side and a chamber side and being movable from the closed state in an opening direction through a normal stroke, within which the main passage is open, and beyond the normal stroke into over-travel, in which the poppet bears on the deflectable ring element and produces a radial deflection of the deflectable ring element while the deflectable ring element remains retained by the core body; and wherein, when the poppet returns from over-travel, the deflectable ring element undergoes shape recovery to its at-rest form and, as it undergoes shape recovery, moves across a surface of the poppet and shears sealant film from that surface.

- The valve core of claim 61, wherein the poppet has a deflecting formation (44) which is spaced from the deflectable ring element in the closed state and throughout the normal stroke and which bears on the deflectable ring element in over-travel so as to produce the radial deflection.
- The valve core of claim 61 or claim 62, wherein the deflectable ring element is a seat ring (30) having a sealing edge (34) on which the sealing face rests in the closed state, and the surface of the poppet across which the seat ring moves is the sealing face.
- The valve core of any one of claims 61 to 63, wherein the deflectable ring element has a wall arranged such that an axial load applied to the deflectable ring element by the poppet in over-travel is converted into the radial deflection, the wall comprising a convoluted wall (32) having one or more convolutions.
- The valve core of any one of claims 61 to 64, wherein the deflectable ring element is formed from a superelastic alloy.
- The valve core of any one of claims 61 to 65, wherein the deflectable ring element is sealed at its periphery so that no inflation gas flows past the deflectable ring element while it is deflected.
- The valve core of any one of claims 61 to 66, wherein the deflectable ring element has a defined wiping edge at an inner surface of the deflectable ring element which is presented to the poppet as the deflectable ring element undergoes shape recovery.
- The valve core of any one of claims 61 to 67, wherein the poppet is displaced from the closed state in response to a pressure differential between the supply side and the chamber side, the poppet beginning the normal stroke at a crack pressure and entering over-travel at a pressure differential exceeding an over-travel threshold higher than the crack pressure.
- An inflation valve for an inflatable chamber, the inflation valve comprising a valve stem (10) having a stem bore (12) and, carried by the valve stem, a deflectable ring element, a poppet (40) and a return spring (50), the poppet having a sealing face (42), a supply side and a chamber side, a main passage (64) past the sealing face being closed by the sealing face in a closed state of the inflation valve, the return spring biasing the poppet towards the closed state, and the poppet being movable from the closed state in an opening direction through a normal stroke and beyond the normal stroke into over-travel; the inflation valve having: the closed state, in which the main passage is closed, the deflectable ring element is in its at-rest form and a purge gap (60) associated with the deflectable ring element is closed; a normal-open state, in which the poppet is displaced from the closed state within the normal stroke, the main passage is open and the purge gap is closed; and a purge state, in which the poppet is in over-travel beyond the normal stroke and produces a radial deflection of the deflectable ring element, the radial deflection opening the purge gap so that a purge flow path (62), separate from the main passage, extends from the supply side through the purge gap to the chamber side; and wherein, when the poppet returns from over-travel, the deflectable ring element undergoes shape recovery to its at-rest form and closes the purge gap.
- The inflation valve of claim 69, wherein the poppet is displaced from the closed state in response to a pressure differential between the supply side and the chamber side, the inflation valve being in the closed state at a pressure differential below a crack pressure, in the normal-open state at a pressure differential at or above the crack pressure and not exceeding an over-travel threshold higher than the crack pressure, and in the purge state at a pressure differential exceeding the over-travel threshold.
- The inflation valve of claim 69 or claim 70, further comprising a core body (20) received in the stem bore, the deflectable ring element, the poppet and the return spring being carried by the valve stem through the core body.

- The inflation valve of claim 71, wherein the core body is threaded into the stem bore and is removable from the valve stem as a unit with the deflectable ring element, the poppet and the return spring.
- The inflation valve of claim 71 or claim 72, wherein the core body, the deflectable ring element, the poppet and the return spring form a valve core according to any one of claims 1 to 68.
- The inflation valve of claim 69 or claim 70, wherein the inflation valve has no separate core body, the valve stem directly carrying the deflectable ring element, which is a seat ring (730), together with the poppet and the return spring, and the valve stem providing a locating formation against which the seat ring is retained and a bore in which the poppet is guided.
- The inflation valve of claim 74, wherein the seat ring has a sealing edge on which the sealing face rests in the closed state and a convoluted wall which converts an axial load applied to the seat ring by the poppet in over-travel into the radial deflection.
- An inflatable chamber comprising a bicycle inner tube or a tubeless bicycle tyre set-up fitted with an inflation valve according to any one of claims 69 to 75.
- The inflatable chamber of claim 76, wherein the inflatable chamber holds liquid sealant.
- A wheel or a bicycle incorporating an inflatable chamber according to claim 76 or claim 77.
- A seat ring (30) for a valve core of an inflation valve, the seat ring being an annular element having a sealing edge (34) at one axial end of the seat ring for seating a sealing face of a poppet, and a wall between the one axial end and an opposite axial end of the seat ring, the wall comprising a convoluted wall (32) having one or more convolutions, the convoluted wall converting an axial load applied to the seat ring into a radial deflection of the seat ring, the seat ring undergoing shape recovery to its at-rest form on removal of the axial load, and the seat ring having a seat ring inner diameter of from 2.0 mm to 2.6 mm and a seat ring wall thickness of from 0.15 mm to 0.30 mm.
- The seat ring of claim 79, wherein the convoluted wall has one convolution or two convolutions.
- The seat ring of claim 79 or claim 80, wherein the convoluted wall has a convolution amplitude of from 0.08 mm to 0.20 mm.
- The seat ring of any one of claims 79 to 81, wherein the seat ring has a seat ring free length of from 2.5 mm to 4.0 mm.
- The seat ring of any one of claims 79 to 82, wherein the seat ring is a continuous ring and the sealing edge is continuous around a circumference of the seat ring.
- The seat ring of any one of claims 79 to 83, wherein the radial deflection is radially outward.
- The seat ring of any one of claims 79 to 84, wherein the seat ring is formed from a superelastic alloy.
- The seat ring of claim 85, wherein the superelastic alloy has an austenite finish temperature below 5 °C.
- The seat ring of claim 85 or claim 86, wherein the superelastic alloy is a nickel-titanium alloy of nominally 55.8 wt% nickel content.
- The seat ring of any one of claims 79 to 84, wherein the seat ring is formed from a resilient polymer.
- The seat ring of claim 88, wherein the resilient polymer is polyether ether ketone.
- The seat ring of any one of claims 79 to 87, wherein the seat ring inner diameter is 2.3 mm, the seat ring wall thickness is 0.20 mm, the seat ring has a seat ring free length of 3.0 mm, and the convoluted wall has a convolution amplitude of 0.12 mm.
- A kit for a valve core of an inflation valve, the kit comprising a seat ring according to any one of claims 79 to 90, a poppet (40) having a sealing face (42) and a purge shoulder (44), and a return spring (50) having a first spring stage (52) and a second spring stage (54), for fitting into a core body (20), the seat ring, the poppet and the return spring being matched so that, when assembled into the core body, the purge shoulder is spaced from the seat ring throughout a normal stroke of the poppet and bears on the seat ring only in over-travel of the poppet beyond the normal stroke, and a purge gap (60) associated with the seat ring is closed in a closed state and in a normal-open state of the valve core.

- A method of operating an inflation valve of an inflatable chamber which holds liquid sealant, the inflation valve comprising a poppet (40) having a supply side, a chamber side and a sealing face (42), a deflectable ring element, a main passage (64) past the sealing face which is closed by the sealing face in a closed state of the inflation valve, and a return spring (50) biasing the poppet towards the closed state, the method comprising, within one inflation event: coupling an inflation device to the supply side, whereby the poppet moves from the closed state through a normal stroke into a normal-open state in which inflation gas flows through the main passage into the inflatable chamber while a purge gap (60) associated with the deflectable ring element remains closed; moving the poppet beyond the normal stroke into over-travel, whereby the poppet produces a radial deflection of the deflectable ring element, the radial deflection opening the purge gap, and inflation gas flows as an air purge along a purge flow path (62), separate from the main passage, from the supply side through the purge gap into the inflatable chamber across surfaces on which sealant film accumulates; and removing the inflation device from the supply side, whereby the poppet returns from over-travel and the deflectable ring element undergoes shape recovery to its at-rest form and closes the purge gap.
- The method of claim 92, wherein the poppet is moved through the normal stroke by raising a pressure differential between the supply side and the chamber side to at or above a crack pressure, and is moved into over-travel by raising the pressure differential above an over-travel threshold higher than the crack pressure.
- The method of claim 92, wherein the poppet is moved into over-travel by a depressor in a head of the inflation device, or in an adaptor fitted between the inflation device and the supply side, on coupling of the inflation device to the supply side.
- The method of any one of claims 92 to 94, wherein, as the deflectable ring element undergoes shape recovery, the deflectable ring element moves across a surface of the poppet and shears sealant film from that surface.
- The method of any one of claims 92 to 95, wherein the air purge carries sealant film from the surfaces on which sealant film accumulates into the inflatable chamber.
- The method of claim 93, or of claim 95 or claim 96 when dependent on claim 93, wherein the crack pressure is from 0.8 bar to 1.5 bar and the over-travel threshold is from 2.5 bar to 5.0 bar.
- The method of claim 93 or claim 97, or of claim 95 or claim 96 when dependent on claim 93, wherein the inflation device is a floor pump and the pressure differential exceeding the over-travel threshold is produced by the floor pump.
- The method of any one of claims 92 to 97, wherein the inflation device is a compressed-gas inflation device.
- The method of any one of claims 92 to 99, wherein the inflatable chamber is a bicycle inner tube or a tubeless bicycle tyre set-up.
- The method of any one of claims 92 to 100, wherein the liquid sealant is an emulsion of latex containing ammonia.
- The method of any one of claims 92 to 101, further comprising repeating the coupling, the moving and the removing over a plurality of inflation events, the purge gap opening and the air purge occurring in each inflation event in which the poppet moves into over-travel.
- The method of any one of claims 92 to 102, further comprising, while the inflatable chamber is pressurised and no inflation device is supplying gas to the supply side, pressing the poppet into over-travel so that the purge gap opens and gas from the inflatable chamber flows through the purge gap from the chamber side to the supply side, carrying sealant film out of the inflation valve.
- The method of any one of claims 92 to 103, wherein the inflation valve comprises a valve core according to any one of claims 1 to 68 or is an inflation valve according to any one of claims 69 to 75.
- A method of manufacturing a seat ring (30) for a valve core of an inflation valve, the method comprising: providing tube stock of a superelastic alloy; laser cutting a profile of the seat ring from the tube stock, the seat ring having a sealing edge (34) at one axial end of the seat ring; forming, in a wall of the seat ring, a convoluted wall (32) having one or more convolutions, the convoluted wall being defined by the laser cutting, by forming on a mandrel, by shape-setting, or by a combination thereof; and shape-setting the seat ring by holding the seat ring in a fixture which constrains the convoluted wall to its intended geometry, heating the

seat ring in the fixture and then water quenching the seat ring, the shape-setting fixing an at-rest geometry of the seat ring to which the seat ring returns by shape recovery, the convoluted wall converting an axial load applied to the seat ring into a radial deflection of the seat ring.

- The method of claim 105, wherein the tube stock has an outer diameter of 2.7 mm and a wall of 0.20 mm.
- The method of claim 105 or claim 106, wherein the tube stock is mounted on a mandrel during the laser cutting.
- The method of any one of claims 105 to 107, wherein the convoluted wall is formed with one convolution or with two convolutions.
- The method of any one of claims 105 to 108, wherein the shape-setting comprises heating the seat ring in the fixture at 500 °C for 10 minutes and then water quenching the seat ring.
- The method of any one of claims 105 to 109, wherein a shape-setting temperature and a shape-setting time are selected so that the seat ring has an austenite finish temperature below 5 °C, and the austenite finish temperature is verified on sampled seat rings after the shape-setting.
- The method of any one of claims 105 to 110, further comprising, after the shape-setting, electropolishing the seat ring to remove surface oxide and a heat-affected edge left by the laser cutting.
- The method of claim 111, wherein a depth of the electropolishing is limited at an inner surface of the seat ring so as to preserve a defined wiping edge formed by the laser cutting.
- The method of any one of claims 105 to 112, wherein the superelastic alloy is a nickel-titanium alloy of nominally 55.8 wt% nickel content.
- The method of any one of claims 105 to 113, wherein the seat ring is a seat ring according to any one of claims 79 to 90.
- The method of any one of claims 105 to 114, further comprising pressing the seat ring into a core body bore (22) of a core body (20) of the valve core until the seat ring seats against an internal shoulder (24) of the core body, with an interference fit between the seat ring and the core body bore.
- The method of claim 115, wherein the interference fit has an interference of approximately 0.03 mm on diameter.
- The method of claim 115 or claim 116, further comprising checking a press-in force or a press-in travel of the seat ring during the pressing.
- The method of any one of claims 115 to 117, further comprising inserting a poppet (40) into the core body so that a poppet stem (41) of the poppet is guided by the core body and a sealing face (42) of the poppet rests on the sealing edge, and fitting a return spring (50) so that the return spring biases the poppet towards a closed state, thereby forming a valve core according to any one of claims 1 to 68.